Named Laws Live at Computable Addresses

Where the NEXUS registry places the latch, the ring oscillator, Griffith's 8, the diffusive floor, no-cloning, Planck, Gödel and the inverse-square law, and what its one empty slot predicts

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Abstract

Named laws live at addresses. Each named result examined here is the value that one of a few root tests returns when it is evaluated inside a class, at a coordinate, and read through a reader: address = (root, class or signature Σ, coordinate θ, reader R). The list covers the latch and Barkhausen loop-gain-1 condition, the secant condition sec(π/N), Griffith's 8, the toggle threshold $a_c(n)$, the resonant rule $2/Q + 1/Q^2$, the diffusive floor e^{π} with its telegrapher sweep, the order bound, no-cloning, Duan–Guo's $1/(1+x)$, Bužek–Hillery's 5/6, the Gisin–Massar surface with Massar–Popescu's $(N+1)/(N+2)$ and Laplace's rule at its edge, Stinespring–Choi dilation, Landauer's $kT\ln 2$, the noise law $\mathrm{CV} \sim \Omega^{-1/2}$, the Planck/Heisenberg–Robertson floor, the diagonal core of Gödel, and the inverse-square flux law. The claim is operational, not rhetorical. One function, given nothing but class descriptors, returns nine thresholds that the literature publishes under different names: five are genuine reproductions (worst relative error 1.3×10^{-11}), one is a correct refusal, two hold by construction and one is a closed form [V]. One Gram inequality returns no-cloning at $p = 1$ and Duan–Guo's rate at its boundary [V]. Robertson–Schrödinger is equivalent to positivity of the 2 × 2 Gram matrix of the deviation vectors, so Planck's floor sits at the same root as no-cloning (10,000 random trials, zero violations) [V]; the author's Law 2, "no matter may hold a location without the potential to be moved from that location", reads as that relation [H]. The resulting registry holds eighteen packages under four roots of different mathematical kind, two verified (loop closure, Gram positivity) and two asserted (dilation, counting measure). It has one slot, local → global, that no registry package resolves, and both corpus families point at it. On the loop side the evidence changed in review. A delayed four-stage ring whose local condition predicts a cycle for τ > 0.1644 shows oscillation at a fixed run length only from 1.25 < τ ≤ 1.5, but at fixed delay the oscillating fraction falls as the run is lengthened: from 37 to 20 of 48 histories at τ = 5, and from 9 to 0 at τ = 2, as T goes from 500 to 4000. The "cycle" is a long transient, as monotone-systems theory requires of a positive loop [V]. Every named result is classical and is cited as such. What is new is the resolver, the root count, the location of the hole, two placements (Planck at the Gram root;

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Laplace's rule at the edge of the cloning surface, for which no published statement was found), and a prediction that review has already partly borne out: the next package the system needs is a global one, and for sign-structured loops it is order structure, which the literature already supplies. Readings that turn constants into coordinates, such as π as the half-turn of the phase circle and *e* as one neper per radian, are marked as hypotheses throughout.

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How to read the status markers

Every claim below carries one of the eight markers that the NEXUS Runtime Specification (v0.1, §7) requires of every NEXUS statement. The specification writes prose definitions for only four of them (SPECIFICATION, IMPLEMENTATION, OBSERVED_RESULT for **[V]**, UNRESOLVED for **[O]**); the definitions for the other four are this paper's working definitions and are marked as such.

Status markers used throughout, as required by the NEXUS Runtime Specification v0.1, §7.

Marker	Runtime name	Meaning in this paper
[V]	VERIFIED	Printed by an executed notebook, check script or corpus file cited in the text. For a literature statement: checked against the source text in this research.
[V-c]	VERIFIED_CONDITIONAL	Verified under a named condition: one class, a finite sample, an optimizer rather than a proof, a chat-only run, or a bibliographic record whose content was not read (working definition).
[R]	RETRACTED	Withdrawn by the programme or by this paper, and kept on the record (working definition).
[F]	FAILED_MEASUREMENT	A prediction or claim that an executed measurement contradicted (working definition).
[O]	OPEN	A named specification gap; stated, not executed.
[S]	SPECIFICATION	A required property, interface contract or intended behaviour.
[I]	IMPLEMENTATION	A concrete executable realization of a specification.
[H]	HYPOTHESIS	An interpretive reading that the executed evidence does not imply (working definition).

Evidence is cited by source. The eight executed notebooks of the loop/channel programme are VRT (The Verb Requirement Test), OB (One Bill, Four Currencies), LOAN (A Loan, Not a Currency), PRICE (The Price of a Crossing), CEIL (A Price or a Ceiling), SIG (The Ceiling Is in the Signature), ARROW (The Arrow and the Coordinate) and REG (The Registry Has Four Roots and One Empty Slot); "REG c3" means cell 3 of REG, counted from zero. "Check 1" to "Check 9" are the placement checks written for this paper (`run_checks.py`, output in `run_checks.out`). "MN" is the merged corpus notebook, cited by line. Appendix C maps every abbreviation to a file.

1. Laws have location because a resolver can compute it

The question this paper answers was put in one sentence: take everything in at once, because "it's about location, where these things actually exist in the space. They are not random, they are all part of a larger complete system. They all have location — even math." The same idea runs through the author's earlier statements. He described package managers as "solutions looking for problems", said that "our universe is the locations of what is here", described science not as a stack but as different software running with different outputs, and on 26 September 2026 asked that the NEXUS work be treated "as a software documentation" in which math laws are libraries reality can pull from **[V as a record of his statements]**. Those are strong claims, and a paper that simply repeats them would add nothing. This paper takes them at their word and asks what they would have to mean for a computer to check them.

"Location" becomes checkable once it means an address that a procedure can resolve. The registry notebook stated the test before running it: "If named results are packages, then one function, given only a class descriptor, should emit them — and each emission should match the value that literature already publishes under its own name" (REG c0). It added a second instruction that turned out to matter more:

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"Count the roots." [V] If the claim were false, the named results would need one special-purpose calculation each, and no small set of tests would cover them. If it were true in its strongest form, one test would emit everything. The evidence lands between those poles. A handful of root tests emit the named results, the tests are of genuinely different mathematical kind, and the system they form has one conspicuous hole. That hole is the most useful thing the scheme has produced, because it says where the next library must go.

The evidence comes from four bodies of work, and they differ in standing. The first is the loop/channel programme: eight notebooks executed on 25 and 26 September 2026 that start from an inverse "verb requirement" test in chemistry and end in a single resolver with four roots and one empty slot. All eight executed cleanly, with sequential execution counts and no error outputs, and every printed number except VRT's is reproduced by a run log [V]. The second is the carry/reader/cohomology family from mid-September: the token-arena programme (§8–§13), its consolidating paper "Transformation without loss", and the adder papers "Two Quotients of One Machine", "Where the Carry Bit Comes From", "What a Cut Must Do to Make a Thing" and "Carry Capacity and the Conditional Cut Law". It supplies the executed evidence for the runtime's meta-operations and the second instance of the empty slot [V]. The third is the constraint-chain cascade, an executed notebook that walks the author's chain from "all things must change" to "there is no end" [V]. The fourth is the specification layer: the NEXUS Runtime Specification v0.1 and its "Reality Mapping" companion. Perplexity wrote both on 26 September 2026 at the author's instruction, and neither is implemented anywhere in the corpus except through `resolve()` [S].

To these the paper adds nine executable placement checks, written to test each placement this paper makes, and eight figures. The checks run in 7.8 s in total. Every check passes; the only FAIL line tests a value that the brief for this paper had wrong ($a_c(2) = 4$ instead of 2), and it is reported as the brief's error below [V]. Literature claims were checked at stated levels: full text where it could be read, bibliographic record otherwise. Where neither was possible, the paper says so.

The argument runs as follows. Section 2 defines the address and the resolver and names the three ways one can move in the address space. Section 3 explains why the programme maps verbs rather than values. Sections 4 to 8 work through the roots: loop closure, which carries half the registry; noise, which turned out to be a loan rather than a currency; the evolution of the taxonomy into the distinction between an arrow and a coordinate; Gram positivity, where no-cloning, the cloning surface and Planck's floor turn out to share one inequality; and the two asserted roots. Section 9 assembles the registry and the runtime it lives in. Section 10 describes the empty slot. Sections 11 to 13 place Gödel through Lawvere, close the constraint chain, and ask what it means for mathematics itself to have location. Sections 14 to 17 position the scheme against prior frameworks, give the complete error log as a first-class result, list what is not new together with every correction, and name the open specifications.

This paper does not claim priority for any individual mathematical result it places. The cyclic-feedback, monotone-systems, stochastic quasi-cycle, quantum-cloning and quantum-channel facts are established, and Section 16 cites them. Its contribution is organizational and operational: a registry that gives each named result an address, a resolver that computes addresses and refuses when a bill cannot be paid, and a class-versus-coordinate test that records whether an extension strictly enlarges what can be done. One section changed after review. A separate session's reading of the draft pointed out that the delayed ring used as evidence for the empty slot is a monotone positive loop, and the test it proposed withdrew the ring's "cycle" (Section 10.1). The correction is kept on the record, as the programme's method requires [V].

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2. An address is a root, a class, a coordinate and a reader

2.1 The definition and the resolver

An address has four slots:

$$\mathcal{A} = (\rho,\ \Sigma,\ \theta,\ R).$$

The root ρ is a test that decides whether a transformation is admissible, and each root is written as one condition. For loop closure the condition is a pair: the phases around the loop must sum to the branch value Φ, and the loop gain must be exactly one.

$$\sum_k \arg H_k(i\omega) + \omega\tau = \Phi, \qquad G\prod_k |H_k(i\omega)| = 1.$$

For Gram positivity the condition is $X^{(1)} - \sqrt{\Gamma}\,X^{(2)}\sqrt{\Gamma} \succcurlyeq 0$, where $X^{(1)}_{ij} = \langle \psi_i \mid \psi_j \rangle$ is the Gram matrix of the inputs and $X^{(2)}$ is the Gram matrix the outputs would need. The class or signature Σ says which transformations are in play: a list of crossings such as three first-order lags, or a lag plus a resonator, or a Copy task with its flags set (exact, deterministic, universal, symmetric, one input). The coordinate θ locates a point inside the class: a decay rate γ, a damping ratio ζ, a line length ℓ, a delay τ, a branch Φ, a stage count N, an overlap x, a success probability p, a copy count (N, M), a spatial dimension d. The reader R is the map from the solution to the number that gets published. It might be the loop gain G_{crit} itself, or the per-stage gain $G_{\text{crit}}^{1/N}$, or the admissibility verdict $\lambda_{\min} \geq 0$, or the fidelity of one output marginal, or the flux through a sphere. These definitions are this paper's specification of the scheme **[S]**.

Write $T_\Sigma(\theta)$ for the set of transformations that class Σ admits at coordinate θ. The resolver is the map from an address to the value at which the root's condition holds with equality, read through R:

$$\text{res}(\mathcal{A}) = R[\,\partial T_\Sigma(\theta)\,].$$

This is the precise sense in which a named law is a boundary: it is the value where the admissible set changes, from damped to oscillating, from monostable to bistable, from clonable to not. In the code, the loop-closure root is **resolve(crossings, tau, Phi)** in REG c1 and the Gram root is **gram_gap(x, p)** in REG c5, followed by a root-finder **[I]**. Appendix A reproduces both. "One resolver" is therefore one interface with two code paths, not one function. No code in the corpus dispatches on root, and a single dispatcher remains a specification **[S]**.

Figure 1 shows three worked addresses. The ring oscillator's secant condition is (loop closure | three first-order lags | Φ = π | per-stage gain). There, $3\arctan(\omega^*/\gamma) = \pi$ forces $\omega^* = \sqrt{3}\,\gamma$ and a total loop gain of 8, which the per-stage reader turns into $\sec(\pi/3) = 2$. No-cloning is (Gram positivity | deterministic copier, p = 1 | overlap x | admissibility $\lambda_{\min} \geq 0$). There, $\lambda_{\min}\big(X^{(1)} - X^{(2)}\big) = -x(1-x)$, which is admissible only at $x \in \{0,1\}$. The diffusive floor is (loop closure | diffusion line with zero decay | Φ = π | loop gain). There, $\ell\sqrt{\omega^*/2} = \pi$ and $\ln G = \ell\sqrt{\omega^*/2}$, which give $e^\pi \approx 23.14$ for every line length. The figure script computes every value with the registry's own code and asserts it against the notebook's saved output **[V]**. One slot in the figure was filled by this paper: the reader for the no-cloning address is the admissibility test that REG c5 prints, since nothing in the corpus names a reader for that address **[V-c]**.

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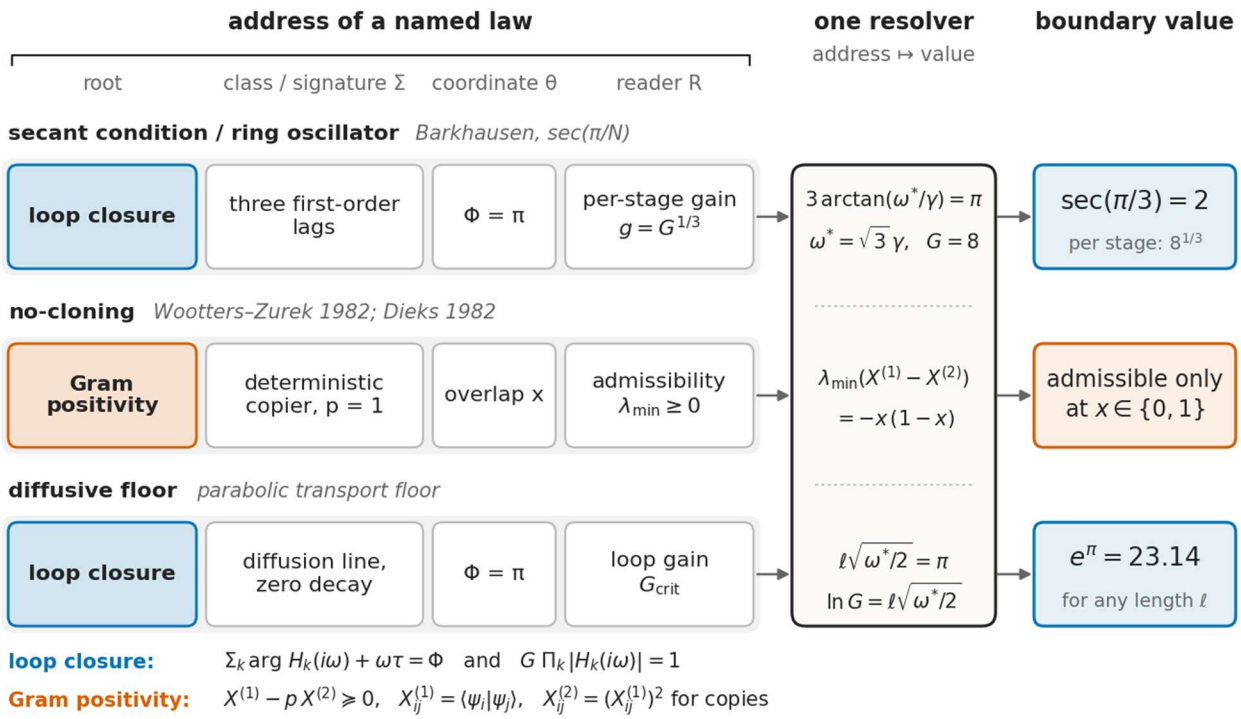


Figure 1. The address of a named law. A named result is the tuple (root, class or signature Σ, coordinate ϑ, reader R). One resolver imposes the root's condition on Σ at ϑ, reads the solution through R, and returns a boundary value. Three worked addresses: the ring-oscillator secant condition (loop closure, three first-order lags, Φ = π, per-stage reader) gives sec(π/3) = 2; no-cloning (Gram positivity, deterministic copier p = 1, overlap x, admissibility reader) is admissible only at x ∈ {0, 1}; and the diffusive floor (loop closure, zero-decay diffusion line, Φ = π, loop-gain reader) gives e^π ≈ 23.14 for any line length.

2.2 Three ways to move

Once laws have addresses, it makes sense to ask what it means to move between them. The corpus distinguishes three kinds of motion, and much of the programme's conceptual work went into keeping them apart [V-c].

A **coordinate move** changes θ inside a fixed class. It is a price: some resource is spent, and the boundary value moves continuously. Lowering the damping of a resonator lowers the critical gain as $4\zeta(1+\zeta)$. Raising a first-order decay rate against diffusion raises it along $\exp\left(\pi\sqrt{1+\mathrm{Da}}\right)$. In ARROW's notation a resource extension leaves the admissible set unchanged, $T_{E(\Sigma)} = T_{\Sigma}$: "same set, different point reached in it" [V-c].

A **class extension** changes Σ itself, so that $T_{E(\Sigma)} \supsetneq T_{\Sigma}$. The programme calls it the arrow. It opens coordinates that did not exist before: admitting delay opens τ, admitting a resonant crossing opens Q, admitting heralded (trace-non-increasing) operations opens a success probability p, and admitting an environment opens its dimension d_a. A boundary that looked absolute inside one class can become a price in the next. The order bound dissolved the moment delay was admitted, and the cloning ceiling of 5/6 turns into a rate as soon as any one of its defining flags is dropped. ARROW's boxed statement is "A class extension opens a coordinate. The coordinate is what you then spend." [V-c]

A **reader change** leaves both class and coordinate alone and changes what is visible. It identifies (takes a quotient) or hides (restricts the interface). The arena programme measured the consequences: symmetry

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hides, since the map $R_{G'} \hookrightarrow R_G$ is injective in every case tested; merging identifies, since that map is injective too; and "only exclusion erases" **[V]**. A reader change can make a law appear or vanish without anything in the underlying set being created or destroyed. It is also the most common way to make a mistake: five of the programme's errors were unintended reader changes (Section 15).

2.3 Why a named law is a boundary, and why every boundary has an address

The registry notebook ends on this sentence: "a named result is a *coordinate bound inside a named class*, and the name records who first compiled that class" (REG c8). The sentence is the notebook's **[V]**; its historical reading of names is interpretation **[H]**. The notebook before it had reached the same point from the other side: "Nothing so far has been a boundary. Everything has been a coordinate exhausted inside a named class" (ARROW c8) **[V-c]**. The two sentences do not contradict each other. ARROW means that no result examined is a boundary absolutely, in every class. Each is the place where a coordinate runs out inside one class. That is exactly why each needs an address: without the class, the boundary is not even defined. The secant condition is a boundary for identical first-order stages without delay. Add delay and the same loop has a new, lower threshold. Make the stages unequal and the true threshold rises above the secant value. A named law, then, is not a free-standing fact about nature. It is the value of a boundary at a particular address, and the address is what makes it reproducible, comparable and extendable.

This also answers the author's word "random". Named results are not scattered, because many of them turn out to be the same address reached by different literatures. Griffith's requirement of a Hill coefficient above 8 for the three-variable Goodwin loop, and the three-stage ring oscillator's per-stage gain of 2, are one address read by two readers: the total loop gain $\sec^3(\pi/3) = 8$ and its cube root **[V]**. No-cloning (1982) and Duan–Guo's probabilistic cloning rate (1998) are "Same matrix. Same inequality. Two names, two literatures, sixteen years apart" (REG c5) **[V]**. That is a unifying presentation, not a discovery: Duan and Guo introduced probabilistic cloning as the generalization of no-cloning, and their p = 1 case reduces to it by construction **[V-c]**. Laplace's rule of succession and Massar–Popescu's optimal estimation fidelity are the same integral (Section 7). The structure is there to be found because the names were attached to boundaries before anyone asked where the boundaries sit.

3. Map verbs, not values: Sun() becomes Sun(a, b, c, d)

The programme began with a remark about method signatures. On 25 September 2026 the author wrote that learning is "taking a method that starts like this Sun() Water() Carbon() and expand that to Sun(a,b,c,d)", that it is the same initial method "that everything interacts with", and that the job was to "map it, not VALUES but the VERBS". The first elaboration, written the same morning, put it this way: "Learning doesn't add values to it. Learning finds arguments that were already callable. What stays fixed while the arguments pile up is the set of verbs, and that set is small." **[S]** The runtime specification later formalized it: "A class extension creates a new callable argument; a resource coordinate selects a value of an argument already available" **[S]**. ARROW, the notebook that measured the difference, wrote it as one line: "The arrow is a class extension. The values of a,b,c,d are resources." (ARROW c8) **[V-c]**

In the language of addresses, Sun() → Sun(a, b, c, d) is a class extension. It adds coordinate slots. Filling a, b, c, d with numbers is a coordinate move, and the numbers the world returns are what a reader reads. "Map verbs, not values" is then a precise instruction: organize knowledge by root and class, the stable part, rather than by the values, which are coordinates and outputs. The address scheme is the Sun() thesis made computable **[H]**.

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3.1 The verb map and what was tested

The verb list was written at the electron level before any notebook existed (session transcript, 25 September 2026, 08:45): nine verbs that a substrate must supply to build logic **[V as a record]**. It was not a result, and the programme treated it as a list of predictions to check. Table 1 gives each verb's meaning in the original wording, the requirement the programme derived from it, and what was actually tested.

***Table 1.** The verb map: each verb's meaning as first written, the requirement derived from it, and what was tested.*

Verb	Meaning as first written	Requirement derived	Tested
Hold	"resist being moved without a price. This is your Law 2"	each state must be actively maintained; species must degrade (VRT R4)	**[O]** in VRT; **[V]** later as conserved-mass thresholds M_c = 3.9771, 2.7708, 1.7189 (OB c16)
Gate	"A's state sets B's permitted transitions" (called "the core verb")	none derived	**[O]** never tested
Invert	"presence gives absence … gating with the sign flipped"	sign of a stage	**[V]** flip one arrow and memory dies (VRT c8); only the sign product matters (OB c6)
Merge	currents meeting at a node "just add (Kirchhoff)"	none derived	**[O]** never tested
Threshold	a nonlinearity; "your Co, manufactured"	Hill exponent $n > 1$	**[V]** boundary exactly at $n = 1$ (VRT c4)
Copy	fan-out; forbidden at the single-quantum level	the no-cloning boundary	**[V-c]** 5/6 at a conjunction of flags; every relaxation priced (CEIL, SIG, ARROW)
Latch	"mutual suppression gives memory … the memory is the loop's shape"	positive loop, $N \geq 2$, loop gain 1	**[V]** (VRT c10; OB c9)
Clock	a relaxation oscillator that turns drift into countable successors	negative loop, $N \geq 3$, $G > \sec(\pi/N)$	**[V-c]** as a Hopf-onset law; the successor reading was never tested
Erase	"Resetting a bit throws away at least kT ln 2"	a minimum reset price	**[O]** stated, never computed

The verb map reaches roots, not values **[H]**. Threshold, Invert, Latch and Clock resolve at root 1 (loop closure). Hold resolves there too through M_c, although M_c never entered the registry. Copy resolves at root 2 (Gram positivity) and, through its ancilla, at root 3 (dilation). Erase is filed under root 4 (counting measure) but has never been computed. Gate, which the first map called "the core verb", and Merge have no address yet. The claim that the verb set is substrate-independent therefore rests on at most seven of nine verbs **[V-c]**. The Clock tested is also not quite the Clock defined. The definition describes a relaxation oscillator with a threshold-and-dump successor map; the tests used smooth Hill rings that lose stability through a Hopf pair, so the programme's clock law is a Hopf-onset law **[V-c]**.

3.2 The Keyes mapping is an analogy with two tested rows

The same chat message mapped R. W. Keyes's criteria for a good logic device onto the author's constraint chain: Co "creates distinction", C1 "requires continuation", C1a "preserves independent continuation", C2 "preserves history through transitions" **[V as a record]**. Table 2 gives the mapping and its standing.

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Table 2. Keyes's device criteria mapped onto the constraint chain, with the standing of each row.

Keyes criterion (as recalled)	Gloss	Mapped to	Standing
Restoration to standard levels	outputs snap to two clean levels	C0, distinction	[V] exercised as Threshold (VRT R1)
Gain	a small input drives a larger output	C1, continuation that does not decay	[V] exercised as the gain criterion (VRT R2)
Input–output isolation	the output cannot write back into the input	C1a, independent channels	[O] no computational counterpart
Feedback/latching	present state carries past state	C2, history preserved	[O] the latch was tested, "history" was not
Dissipation on erase	resetting costs energy	"price to prove change"	[O] not computed

Keyes's paper is verified only at the level of its bibliographic record: *Science* 230(4722), 138 (1985), DOI 10.1126/science.230.4722.138 (**Science**; **PubMed**) [V-c]. Its exact list of criteria could not be read, so the left column is a recollection and should not be quoted as Keyes's words [O]. The planning note behind the table also listed fan-out and nonlinearity, which got no row. The mapping stands as an analogy, and only its first two rows have been tested [V-c].

3.3 The inverse test: predictions written before the chemistry

VRT asked the verb map to predict, "written before touching the chemistry", what a chemical reaction network must contain in order to latch and what it must contain in order to count (VRT c0) [V]. The substrate had "no voltage, no rails, no channel, no supply", and the notebook promised that "Nothing below uses any knowledge of the genetic toggle switch or the repressilator." Five predictions followed.

- **R1 Threshold.** Mass action cannot latch at any production rate. At $a = 10^6$ the network has one steady state for n = 0.5, 0.9, 0.99, 0.999 and 1.0, and three for n = 1.001, 1.01, 1.1 and 2.0 [V].
- **R2 Gain.** A critical production rate exists. Onset occurs where $|f'(u^*)| = 1$, which gives $u^{*n} = 1/(n-1)$ and $a_c(n) = n(n-1)^{-(n+1)/n}$. The formula held in 9 of 9 cases, with one fixed point at $0.98\,a_c$, three at $1.02\,a_c$, and round-trip gain 1.00000000 at onset [V]. "R1 and R2 are not two requirements, they are one requirement seen from two sides."
- **R3 Positive loop.** Flip one arrow and three steady states become one attractor, (0.74, 3.538). "Same gain. Same threshold. Same dimension. Same degradation. One arrow flipped. Memory is gone." [V]
- **R4 Hold** was never tested in VRT [O]. It was exercised later, as conserved mass in OB.
- **R5a Parity** (even rings latch, odd rings clock) scored 17 of 18 raw [V-c]. The one miss, N = 3 with a = 1000 and n = 2.0, sat at per-stage gain 1.9801, below 2, and it produced the sharper law: oscillation occurs if and only if $G > \sec(\pi/N)$, confirmed 5 of 5 at $a = 10^7$ [V]. VRT's scoring line labels the overall 17 of 18 as the "clock side", which is a mislabel [R as wording].
- **R5b Dimension cost.** The claim that counting costs one more dimension than memory was refuted as stated by OB, where a single delayed species clocks in 6 of 6 cases [F]. PRICE re-scoped it to total dynamical order $p \geq 3$ in the delay-free class.

The boxed result was "memory = positive loop, N≥2; counting = negative loop, N≥3, G>sec(π/N)", with "One requirement underneath both — round-trip gain >1" (VRT c17) [V-c]. None of it is new: the toggle switch and the repressilator are the biological instances, and the secant condition is textbook. Its value was

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methodological. A verb list written for electrons predicted, before any simulation, the requirement structure of a chemical network, and its one failure produced a sharper law.

4. Root 1: loop closure carries half the registry

4.1 One characteristic equation for every ring

Consider any cyclic network $\dot{x}_i = f_i(x_{i-1}) - \gamma_i x_i$, with signed stage gains g_i and signed loop product $P = \prod_i g_i$. Linearized about a steady state, its characteristic equation is

$$\prod_i (\lambda + \gamma_i) = P,$$

and where the gains sit around the ring makes no difference (OB c0) **[V]**. At a Hopf crossing $\lambda = i\omega$, the equation splits into a phase condition, $\sum_i \arctan(\omega/\gamma_i) = \pi$, and a modulus condition, $\prod_i \sqrt{\omega^2 + \gamma_i^2} = |P|$.

OB tested the equation against the eigenvalues of 3000 random cyclic matrices (N from 3 to 11, decay rates in (0.05, 6), mixed signs), and all 3000 agreed (OB c6) **[V]**. Moving gain from one stage to another, as in (2, 2, 2), (8, 1, 1), (4, 2, 1) or (0.5, 4, 4) at γ = 1, changes nothing: each product is 8, each critical value is 8, and each ring clocks **[V]**.

The nonlinear test ran 30 asymmetric mixed-sign rings whose steady-state residual was checked (at most 1.9×10^{-13}), excluding a knife-edge band around the threshold. All 30 agreed **[V-c]**. OB's scoring table claims "40/40". That count went stale when a trim changed the cell's target from 40 to 30, and it is corrected here to the printed 30 of 30 **[R as wording]**.

The notebook says of itself that "a control theorist would recognize every line of Step 1 as the Nyquist criterion" (OB c17) **[V]**. The phase-and-magnitude pair is Nyquist's marginal case (**Nyquist 1932**) **[V-c]**. It is also Barkhausen's self-excitation condition: loop gain of unit magnitude and total phase zero mod 2π, in the positive-feedback convention (**Barkhausen stability criterion**). That secondary source stresses that the criterion is "a necessary condition for oscillation but not a sufficient condition" **[V]**. The resolver uses the negative-feedback convention (total lag = Φ, with Φ = π for a negative loop), and a registry should state its convention **[S]**.

For identical stages the phase condition gives $\omega^* = \gamma\tan(\pi/N)$ and a per-stage gain of $\sec(\pi/N)$: 2 at N = 3, 1.236068 at N = 5, 1.109916 at N = 7, falling toward 1 as N grows (OB c10) **[V]**. This is the secant condition of Tyson and Othmer (1978) and Thron (1991). Sontag states it as $\sigma_1 \cdots \sigma_n/(\lambda_1 \cdots \lambda_n) < \sec(\pi/n)^n$, with the bound equal to 8 at n = 3, 4 at n = 4 and about 2.88 at n = 5, and "achieved exactly when all the λi's are the same" (**Sontag 2006**) **[V]**.

4.2 The master phase-closure law and its three corners

With equal decay and equal per-stage gain G, one first-order stage at the crossing contributes phase $\arctan\sqrt{G^2 - 1} = \arccos(1/G)$ and a delay τ contributes $\tau\sqrt{G^2 - 1}$. The whole family then obeys one law (OB c9) **[V]**:

$$N\arccos(1/G) + \tau\sqrt{G^2 - 1} = \Phi.$$

The branch Φ is set by the sign of the loop, and each branch is a corner of the law (OB c9, c15) **[V]**. Figure 2 draws the law for one to four stages.

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Φ = 0 is the latch. A real eigenvalue crosses at ω = 0, which is available only to a positive loop. At ω = 0 the law collapses to $\prod_i (G_i/\gamma_i) = 1$: the loop-gain-1 saddle-node condition of a cross-coupled latch or SRAM cell. The notebook's sentence is "Memory is the zero-frequency corner of the clock equation." [V] The toggle threshold $a_c(n)$ is this corner mapped into Hill parameters. At the symmetric steady state the loop gain is $[nu^n/(1+u^n)]^2$, and setting it to 1 on the steady-state map $u(1+u^n) = a$ returns $a = 2.000000000000$ for n = 2 (Check 1) [V]. Check 2 confirms $a_c(2) = 2$, $a_c(3) = 1.1905507890$ and $a_c(4) = 1.0131142475$ by fixed-point counting and 24 simulations per setting [V]. For n = 2 the result is also exact without the general formula. Subtracting the two steady-state equations gives $(x-y)(1-xy) = 0$, so the asymmetric states are $x, y = \left(a \mp \sqrt{a^2-4}\right)/2$, which are real only for $a > 2$. At a = 3 they are (0.3820, 2.6180), matching the simulated end states [V]. The brief for this paper had asked for $a_c(2) = 4$. That value is wrong: the toggle is already bistable at a = 2.02, 2.1, 3.0 and 3.9, and 4 is only the $a \to \infty$ supremum of the n = 2 loop gain, n^2, not a bifurcation value. Check 2 records the requested value as a FAIL line [F]. The registry labels the formula "Cherry–Adler". This paper did not check that attribution against the original paper, and it should be verified before print [O].

Φ = π is the clock. A complex pair crosses on a negative loop, and this is the branch of the secant condition. With delay, one stage can pay what three stages pay without it. At G = 2, stages alone suffice for N = 3; N = 2 needs τ > 0.604600 and N = 1 needs τ > 1.209200 (OB c10) [V]. The exchange rate along the law is $d\tau/dN = -\arccos(1/G)/\sqrt{G^2-1}$: 0.752275 at G = 1.5, 0.604600 at G = 2, 0.435210 at G = 3, and exactly $\pi/(3\sqrt{3}) = 0.604599788078$ at G = 2 (Check 2) [V]. A single self-repressing species with delay clocks where the law says it should, 6 of 6 across τ = 0.8, 2.0 and 5.0 (OB c12) [V]. This is the classical result for $x' = -ax + bx(t-\tau)$: the Hopf crossing of the delayed negative loop occurs at $\tau_c = \arccos(-a/c)/\sqrt{c^2-a^2}$, which is the master law at N = 1 (**Hayes 1950**; the statement is recalled from secondary texts) [V-c]. It is also the mechanism of delay-induced oscillators in gene circuits (**Lewis 2003**; **Monk 2003**) [V-c].

Φ = 2π is preempted. A positive loop can also meet the phase condition with a complex pair at total lag 2π. The law gives finite thresholds for N ≥ 5 without delay: 3.236067977 at N = 5, 2.000000000 at N = 6, 1.414213562 at N = 8, each equal to $\sec(2\pi/N)$ [V]. OB predicted that a two-repressor ring would therefore become a clock "given enough delay". It did not. Every such run latched: at N = 2, τ = 2.0, G = 6.0 the endpoints were (0.0, 8.352) and (8.352, 0.0), with swing 0 [F]. The refutation produced what the notebook calls its best result, the preemption argument. A real crossing opens at loop gain 1. Every complex crossing needs ω > 0 and therefore costs modulus. So the latch always opens first, and the law "is exact about the stability of the state it is evaluated at, and silent about attractors elsewhere" (OB c15) [V]. At N = 4, τ = 3.0, G = 5.0 OB saw a sustained oscillation beside the latch at its run length [V]. Section 10.1 shows that such oscillations die when the run is lengthened, so the "coexisting cycle" was a long transient [R]. It is where the empty slot of Section 10 was first sighted, and where it was later corrected.

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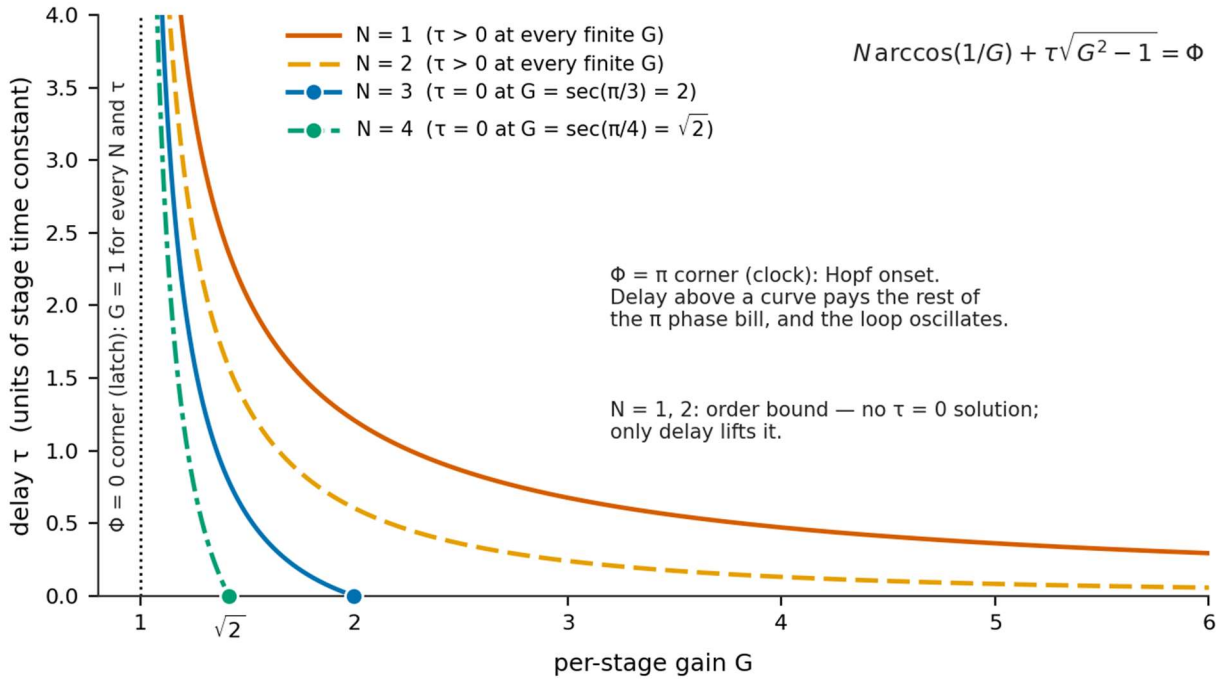


Figure 2. The phase-closure law with delay. For N identical first-order stages (time constant 1) and a pure delay τ, the Hopf boundary at Φ = π satisfies N arccos(1/G) + τ√(G² − 1) = Φ; the curves give the critical delay τ_crit(G) for N = 1–4. Without delay the boundary exists only for N ≥ 3, at G = sec(π/N) (2 for N = 3, √2 for N = 4); for N = 1 and 2 only delay can pay the remaining phase. The Φ = 0 (latch) corner collapses to G = 1 for every N and τ.

4.3 The transport axis: what one radian of phase costs

The law above covers only first-order stages and pure delay. PRICE and CEIL asked whether the kind of crossing changes the bill, and the answer organizes every loop threshold along one axis. Each crossing kind charges a different magnitude price for the phase it supplies **[V-c as an organizing reading of executed results]**.

Lags pay phase with attenuation, and no lag supplies more than π/2. **Delay** pays phase at no magnitude cost; it is the canonical all-pass element, which is why delay drives the critical gain down toward 1 and no further. For a two-lag loop, τ = 0.3 gives 7.3463, τ = 1 gives 2.7071, τ = 5 gives 1.2087 and τ = 100 gives 1.0009 (PRICE c7) **[V]**.

Resonance pays phase and returns magnitude. A lag closed through a second-order resonator of damping ζ, tuned to the lag's rate, crosses at $\omega^* = \sqrt{1 + 2\zeta}$ with

$$G_{\text{crit}} = 4\zeta(1 + \zeta) = \frac{2}{Q} + \frac{1}{Q^2}, \qquad Q = \frac{1}{2\zeta},$$

confirmed 10 of 10 from ζ = 1 (where it returns the three-lag value 8) down to ζ = 0.001 (PRICE c5) **[V]**. A first derivation gave $4\zeta = 2/Q$. It was off by 20% at ζ = 0.2 and was replaced by the exact form **[R]**. At ζ = 0.01 the resonator's own magnitude at ω^* is 35.18, so the external gain needed falls to 0.0404. The review that caught the loose wording insisted that "the resonator supplies the gain" must be paired with "by temporarily returning energy previously stored in its reactive degrees of freedom", and CEIL accepted it: "a passive resonator creates nothing" **[V]**. The Routh–Hurwitz form for a detuned resonator is $2\zeta(\rho + 1/\rho) + 4\zeta^2$, with

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ρ the ratio of the resonator's frequency to the lag's rate, so the registry value is the minimum over detuning (literature check, derived and checked numerically) [V-c].

Diffusion is where the axis ends. Through a diffusion line $H = e^{-\ell\sqrt{s}}$, one has $\sqrt{i\omega} = \sqrt{\omega/2}\,(1+i)$, so attenuation in nepers equals phase lag in radians at every frequency. Check 4 confirmed the identity across 4001 frequencies to 3×10^{-12} [V]. Phase π therefore costs a factor e^{π} in magnitude, and the loop threshold is

$$G_{\text{crit}} = e^{\pi} = 23.140692633$$

for every line length. The crossing frequency moves as $2\pi^2/\ell^2$ while the threshold stays fixed: ℓ from 0.01 to 100 gives 23.140692633 every time (CEIL c3) [V]. An exact finite domain returns the same value to twelve digits, and a closed-loop PDE with a saturating nonlinearity oscillates at $1.03\,e^{\pi}$ and decays at $0.97\,e^{\pi}$ (CEIL c7) [V]. With first-order decay the threshold rises along $\exp\left(\pi\sqrt{1+\mathrm{Da}}\right)$: 15 of 15 in CEIL and 24 of 24 in Check 4, to 1.8×10^{-15} [V].

The **telegrapher line** $e^{-L\sqrt{s(s+2a)}/c}$ connects the two ends of the axis. As damping rises the threshold sweeps continuously from pure delay to diffusion: 1.000000 at a = 0, 1.638507 at a = 0.5, 5.403934 at a = 2, 20.028736 at a = 10, and 23.140693 at $a = 10^6$ (CEIL c5) [V]. Figure 3 collects the four crossing classes on one axis.

The structural reason for the two ends is classical [V-c]. A diffusion line is minimum-phase, so Bode's gain–phase relation ties its phase to its attenuation (**Bode 1940**). A delay is all-pass, so it reaches –π at zero attenuation. The literature check found no source that states the e^{π} threshold, the decay-corrected form, or the telegrapher sweep. The ingredients are all classical, from Kelvin's cable to Ångström's thermal waves. The closest precedents located are a 1978 paper on distributed-RC phase-shift oscillators, which could not be read, and Busenberg and Mahaffy's repression loop closed through diffusion (**Busenberg & Mahaffy 1985**). That search was at title level, so the absence is not evidence of novelty [V-c].

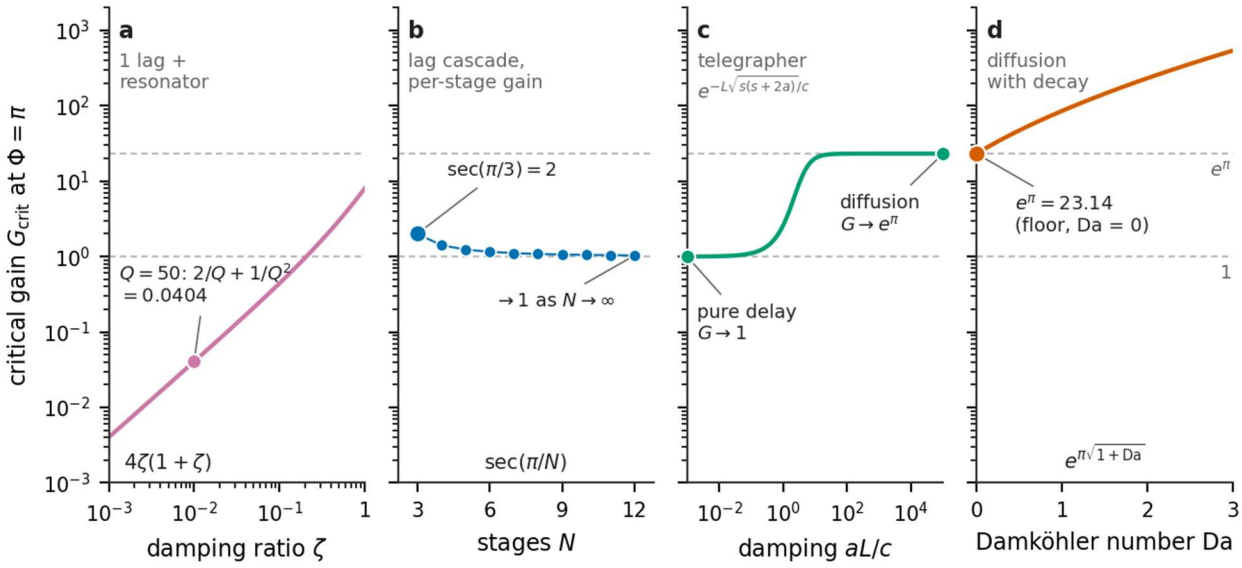


Figure 3. One transport axis for the Φ = π (clock) boundary. Critical loop gain from the same resolver for four crossing classes: (a) one lag plus a resonator, G_crit = 4ζ(1+ζ) = 2/Q + 1/Q² (0.0404 at Q = 50); (b) N identical lags, per-stage gain sec(π/N), which is 2 at N = 3 and tends to 1; (c) the telegrapher line, moving continuously from pure delay (G → 1) to

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diffusion (G → e^π) as damping aL/c grows; (d) diffusion with decay, G = e^{π√(1+Da)}, whose floor at Da = 0 is e^π ≈ 23.14. Dashed lines mark G = 1 and G = e^π. Panel (b) plots per-stage gain; the others plot total loop gain.

The **order bound** comes out of the same axis. Delay-free loops built from lags and resonators have no finite Hopf threshold at total dynamical order 2, at any gain up to 10^9 and even for a resonator with ζ = 0.01 (PRICE c3) **[V]**. Every order-3 arrangement tested has one, and at fixed order 3 the threshold ranges from 8.0 down to 0.0404, "a factor of 198" **[V]**. LOAN had predicted that a single second-order stage with a negative loop would oscillate. PRICE refuted that prediction **[F]**, and the lesson was to count total order, not stages.

4.4 Nine thresholds from one function, and which of them count

REG handed nine class descriptors to `resolve()` and compared each output with the value the literature publishes. Check 1 re-ran the table, added an independent residual $\left|G\,H(i\omega^*) - e^{-i\Phi}\right|$ computed from a separately coded transfer function, and printed the crossing frequencies **[V]**. Table 3 shows the result.

***Table 3.** Nine named thresholds from one resolver, and which of them count as reproductions.*

Row	Class descriptor	Φ	Emitted	Published value (name)	Standing
1	2 lags	0	1.000000	loop gain 1 (saddle-node, SRAM latch)	holds by construction
2	Hill toggle, n = 2	0	2.000000	$a_c(n)$ (attributed to Cherry–Adler)	closed form, not routed through `resolve()`
3	3 lags	π	2.000000 per stage (loop 8)	$\sec(\pi/3)$ (Barkhausen, ring oscillator)	reproduction; ω* = 1.7320508
4	5 lags	π	1.236068 per stage	$\sec(\pi/5)$	reproduction; ω* = 0.72654253
5	lag + resonator, ζ = 0.01	π	0.040400	$2/Q + 1/Q^2$ (crystal-oscillator design rule)	reproduction; ω* = 1.0099505
6	diffusion line, zero decay	π	23.140693	e^{π} (parabolic transport floor)	reproduction; ω* = 2π²
7	telegrapher, a = 0	π	1.000000	G → 1 (pure delay)	holds by construction
8	telegrapher, a = 10⁶	π	23.140693	e^{π} (same axis, far end)	reproduction; relative error 1.3e-11
9	2 lags	π	none	no finite Hopf (order bound)	correct refusal

The standing column matters more than the printed "9 of 9". Row 1 holds by construction: a unit lag has DC gain 1, so any cascade of them returns exactly 1 on the Φ = 0 branch. Row 7 is forced the same way, since a lossless delay has unit magnitude. Row 2 is a closed form evaluated inline; it reaches `resolve()` only through the toggle map outside it, which returns a = 2.000000000000. Row 9's match is a match of absences. It is backed by an independent check that the roots of $(1+s)^2 + G$ keep real part −1 for G from 10^{-3} to 10^8, and by the observation that adding τ = 0.1 to the same class produces a finite G_{crit} = 20.671069, so the bound is class-relative **[V]**. Rows 3, 4, 5, 6 and 8 are the genuine reproductions: root-finding alone lands on the textbook closed forms. Their worst relative error is 1.3×10^{-11}, in row 8, where the telegrapher at finite damping approaches e^{π} from below, and their worst loop residual is 2.4×10^{-12} **[V]**. The honest count is five reproductions, one correct refusal, two identities and one closed form. REG's own sentence still holds: "A registry that only ever says *yes* is a lookup table; this one refuses." **[V]**

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4.5 Four corrections the placement forced

Placing each result at an exact address exposed four places where the usual statement is loose.

The secant condition is tight only for identical stages. Sontag's statement makes $\prod g_i < \sec^N(\pi/N)$ a sufficient stability condition for arbitrary rates, and exact only when all rates are equal (**Sontag 2006**) **[V]**. With unequal time constants the real threshold is higher. So "sec(π/N)" lives at the address (loop closure | N identical first-order stages | Φ = π | per-stage gain), not at "any N-stage ring" **[V]**.

Griffith's 8 is an infimum, approached only as a → ∞. In the three-variable Goodwin loop the loop gain at the steady state is $K = nu^n/(1+u^n) < n$, and the Hopf point is K = 8 exactly, with onset eigenvalues −3 and $\pm i\sqrt{3}$ (Check 3) **[V]**. The tested Hill coefficients up to 8 (n = 2, 4, 6, 7.9 and 8) never oscillate at any of 301 values of a in $[10^{-3}, 10^{12}]$. The smallest oscillating coefficient at a given a falls toward 8 from above: 16.000 at a = 2, 9.131 at a = 10, 8.134 at a = 100, and 8.000000000172 at $a = 10^{12}$ **[V]**. Asymptotically $n_{\min}(a) - 8 \approx 8a^{-8/9}$, which matches the printed values to all printed digits from $a = 10^3$ upward (evaluation for this paper) **[V-c]**. The n = 8 case passes on a margin of -7.181×10^{-1} that shrinks as $-1/(3a^{8/9})$, so its PASS depends on where the grid stops **[V-c]**. The printed side remark "a ≤ 1 never oscillates" is untested and false in a far corner: $a_H(n)$ dips below 1 for $n \gtrsim 2.39 \times 10^4$, with a minimum near 0.9999846 **[R]**. Griffith's result is verified here only through its record, not its text (**Griffith 1968**; **Goodwin 1965**) **[V-c]**. Sontag confirms that $\sec^3(\pi/3) = 8$ **[V]**.

The diffusive form needs the π²-normalized Damköhler number. The identity $G_{\text{crit}} = \exp(\pi\sqrt{1+\text{Da}})$ holds only with $\text{Da} = \gamma L^2/(D\pi^2) = kT/\pi^2$, where $T = L^2/D$. With the conventional $\text{Da} = kT$ the same threshold reads $\exp(\sqrt{\pi^2 + kT})$ **[V]**. Mixing the conventions is not a small error. At kT = 1 the true threshold is 27.028944477, while the misapplied form gives 85.019695223. At kT = 10 the numbers are 86.274445654 and 33506.143065592 **[V]**.

The order bound depends on the class. "Order ≥ 3" holds for delay-free loops of lags and resonators in which no stage activates itself. With every diagonal Jacobian entry negative, a two-stage negative loop has negative trace and positive determinant at every gain **[V-c]**. Delay dissolves the bound, as the τ = 0.1 line above shows **[V]**. Two-variable oscillators with self-activation exist, so the hypothesis must be stated **[V-c]**. PRICE first called the bound a gate that "no amount of anything else substitutes". The same notebook's delay runs contradict that wording **[F]**, and CEIL recorded that the gate held "only *within* the delay-free rational class" **[V]**.

One loop-side quantity has no closed form and no registry entry. This is the conserved-mass threshold of a toggle whose two species draw on a shared pool: M_c = 3.9771, 2.7708 and 1.7189 for (n, k) = (2, 1), (3, 1) and (4, 5), found by bisection (OB c16) **[V]**. It is Hold made measurable. It is absent from REG's eighteen packages and remains an address without a resolver **[O]**.

5. Noise is a loan, not a currency

If loop closure prices every threshold, where does noise go? LOAN posed the question in a form that could fail: "Is noise a fifth currency … or is it a loan that appears as oscillation at small scale and disappears as the substrate grows?" (LOAN co) **[V]**. The two readings were given operational signatures before any simulation ran. A currency would keep the coefficient of variation (CV) constant as system size Ω grows, would buy coherence, would introduce a new frequency, and would smear the threshold away. A loan would make CV

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fall as $\Omega^{-1/2}$, pin the decorrelation rate at the deterministic damping $|\mathrm{Re}\,\lambda|$, keep the frequency at $\mathrm{Im}\,\lambda$, and leave the threshold locatable at $\sec(\pi/N)$.

The test ran a three-repressor ring as a chemical master equation by tau-leaping. It measured the circulant order parameter $z = \sum_k x_k\, e^{2\pi i k/3}$, took the decorrelation rate Γ and the frequency from the FFT autocorrelation of z, and computed the reference eigenvalues $\lambda = (G/2 - 1) \pm iG\sqrt{3}/2$ in advance (LOAN c1) **[V]**. Below threshold, at G = 1.70, CV falls from 0.2715 at Ω = 30 to 0.0154 at Ω = 10,000. The log-log slope is −0.497. Over the same range Γ is flat (slope −0.008) near $|\mathrm{Re}\,\lambda|$ = 0.150, and the frequency converges to 1.475 against a predicted 1.4722 **[V]**. Above threshold, at G = 2.30, the pattern inverts. CV barely moves, from 0.5698 to 0.5135 (slope −0.015), and Γ falls with slope −0.893, from 0.0886 to 0.0006 **[V]**; that is the signature of a true limit cycle, whose phase diffuses more slowly as the system grows **[V-c]**. A blind sweep, which was not told where the threshold is, compared Γ at Ω = 100 and Ω = 3000. The ratio was 1.03 at G = 1.8, 1.55 at G = 1.9, 4.62 at G = 2.0 and 32.28 at G = 2.5, which recovers the threshold between 1.9 and 2.0 against $\sec(\pi/3) = 2$ (LOAN c5) **[V]**.

The verdict: "Noise is a loan, not a currency … amplitude on credit, repaid at the damping rate", and "the law does not degrade below threshold" (LOAN c6) **[V]**. Of the two signatures, the flat Γ is the stronger. A falling CV alone would also fit a sub-threshold system that merely rings, but a decorrelation rate pinned at $|\mathrm{Re}\,\lambda|$ ties the noisy dynamics to the deterministic eigenvalue **[V-c]**. None of this is new. The $\Omega^{-1/2}$ scaling is the system-size expansion (**van Kampen 1961, record**). Noise-sustained "quasi-cycles" whose coherence is set by the real part of the eigenvalue are McKane and Newman's resonant amplification (**McKane & Newman 2005**). The simulation method is Gillespie's (**1977**; tau-leaping **2001**) **[V-c]**.

In address terms, the noise law lives at (counting measure | finite-copy master equation on a ring | system size Ω | CV of one species), and the registry files it under root 4 **[V-c]**. The "loan" reading is interpretive. Ω is a coordinate that borrows amplitude at small sizes and repays it at the damping rate, and it buys no coherence **[H]**. The evidence is one system only. No latch, delayed loop or conserved-mass network was run under noise, and the width of the blind crossover was stated to narrow with Ω but never measured **[O]**.

6. From currencies to arrows: how the taxonomy found its final form

The address scheme's three kinds of motion were not assumed. The programme found them by trying and discarding five vocabularies in five notebooks. The history matters, because the older words still appear in the notebooks and mean different things in each. Table 4 gives the sequence. Each revision after OB followed an external review uploaded between notebooks, and the revised text often reuses the review's wording. LOAN's boxed statement closely follows review 113d86ac; SIG's sentence on polymorphism follows review 1b2d726a; ARROW's definitions of T_Σ follow review 379599a2. The reviews deserve the credit **[V]**.

Table 4. How the taxonomy found its final form, notebook by notebook.

Notebook	Vocabulary	What it claimed	What became of it
OB	one bill, four currencies	"One conserved bill … Four currencies … interchangeable ways of paying one bill"	**[R]** "conserved" and "interchangeable" withdrawn in LOAN; only stages ↔ delay was ever executed

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Notebook	Vocabulary	What it claimed	What became of it
LOAN	phase-closure requirement; loan	"Φ is a phase-closure requirement, not a conserved quantity"; "the resources are not peers. Only two genuinely interchange"	[V] kept
PRICE	gate vs price	order ≥ 3 is a gate "no amount of anything else substitutes"	[F] delay dissolves it in the same notebook
CEIL	price, floor, ceiling by whether a resource "appears in its statement"	diffusion is "the first floor"; Copy is a ceiling	[R] demoted in SIG to "a discovery heuristic"
SIG	invariance under resource extension; ceiling as conjunction	"A ceiling is a conjunction. Name the conjuncts and each one has a rate."	[V] kept; "no larger class to escape into" [R] withdrawn
ARROW	class extension (arrow) vs resource coordinate	$T_{E(\Sigma)} \supsetneq T_\Sigma$ opens a coordinate; $T_{E(\Sigma)} = T_\Sigma$ spends one	[V-c] measured for ancilla and postselection only
after ARROW	Choi-rank refinement	d_a from 1 to 2 is a class extension; beyond the Choi rank it is inert	[V-c] six frontier points; never written back into ARROW
REG	the same criterion at the global layer	existence of an attractor is a threshold; its basin is a coordinate	[R] the example was a long transient (Section 10.1); the attractor-set criterion is [O]

SIG found that the Copy "ceiling" was not a property of the verb but of its specification. **Copy()** is not forbidden; **Copy(exact, deterministic, universal, symmetric, inputs=1)** is, and "every argument you unset opens a lawful realization with a published price" (SIG c12) [V]. Holding every flag except exactness gives the symmetric optimum 5/6, which no ancilla dimension moves. Dropping universality makes copying free for a known orthogonal pair: CNOT copies $|0\rangle$ and $|1\rangle$ with error 0.00 and fails on their superposition with error 0.7654 (SIG c3) [V]. Dropping determinism buys exactness at success probability $1/(1+x)$. Dropping symmetry moves along a frontier that reaches $(F_1, F_2) = (1.000000, 0.499999)$. Dropping "one input" and reading the estimation edge gives $(N+1)/(N+2)$, which climbs toward 1 (SIG c5–c9) [V]. That value is the state-estimation fidelity, the $M \to \infty$ edge of Section 7.3, not the fidelity of an $N \to N+1$ cloner, which is $((N+1)^2 + N)/\big((N+1)(N+2)\big)$; the two tasks should not share a label [V]. Two internal inconsistencies remain on the record. SIG co says 5/6 holds five flags, but its own c11 and c12 show that exactness is released, so the correct count is four [R as wording]. SIG calls each relaxation "a resource extension", while ARROW calls each "a class extension away" [R as wording]. This paper follows ARROW.

ARROW made the distinction measurable. If an extension enlarges the admissible set, the achievable (F_1, F_2) frontier must move outward; if it only spends a coordinate, the frontier stays put (ARROW co) [V]. Raising the ancilla dimension from 1 to 2 moved the frontier by 4.35×10^{-2}. Raising it from 2 to 3 moved it by 1.95×10^{-5}, which is optimizer noise. The Choi rank of the optimal channel is 1 at $d_a = 1$ and 2 at d_a = 2, 3 and 4 (ARROW c3, c5) [V]. Postselection enlarges the set: an explicit single-Kraus cloner reaches $p = 1/(1+x)$ exactly, 7 of 7 to twelve digits, but it is trace-preserving only at x = 0. At x = 0.3, for instance, the eigenvalues of $\mathrm{Tr}_{\mathrm{out}} J$ are (0.64497, 1): trace-non-increasing, not trace-preserving. ARROW's label was "Class extension, measured." (ARROW c7) [V].

A review uploaded after ARROW caught a real inconsistency, and the session accepted it. By ARROW's own criterion, d_a from 1 to 2 *is* a class extension, since $T_{d_a=1} \subsetneq T_{d_a=2}$. Only $d_a \geq 2$ is an inert coordinate, and "saturation point = Choi rank" was over-claimed, because rank had been measured at the optimum and not across the set [V-c]. The corrected statement: for the tested cloning frontier, the first complete attainable

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set occurs at $d_a = 2$, matching the rank-2 optimal channel. It was checked at six frontier points, all rank 2, with a maximum gap between $d_a = 2$ and $d_a = 3$ of 1.40×10^{-5} (setcomp.py) [V-c]. The general form, $d_{\min}(\mathcal{F}) = \max_{x \in \mathcal{F}} \operatorname{rank} J_x$, would be a set-completion theorem and is unproved [O]. ARROW's cell 5 still prints "Saturation point = Choi rank", and the runtime's claims ledger still lists the Choi-rank statement as [V]. Both should read [V-c] [R as wording].

REG then claimed the same distinction one level up. For the delayed ring of Section 10 it wrote "**Existence** of the cycle is a threshold; its **basin** is a continuous coordinate" (REG c8). The horizon test of Section 10.1 withdraws the example: what REG measured was the share of histories still oscillating at a fixed run length, which falls as the run grows, not a basin [R]. The proposed criterion for the global layer is: "If A_Σ is the set of attractors a class admits, then a resource extension leaves A_Σ unchanged while moving basins, and a class extension adds an attractor *type*." It has not been tested [O].

The lexicon this paper uses is therefore ARROW's, with the Choi-rank refinement. A **price** is a coordinate move inside a fixed class. A **floor** is a coordinate exhausted inside a class that a class extension can escape: e^{π} is a floor within parabolic transport, and the telegrapher class escapes it. A **ceiling** is a coordinate exhausted inside a specification whose conjuncts are each a class extension away: 5/6 within (deterministic, universal, symmetric, one input). ARROW's own summary was "Three revisions of one concept is not three discoveries; it is a measure of how loose the first version was." [V] The count was five vocabularies before ARROW and two refinements after it, which makes that sentence truer than the notebook knew.

7. Root 2: one Gram inequality holds no-cloning, the cloning surface and Planck's floor

7.1 No-cloning is the p = 1 corner

A unitary cannot change inner products, so it cannot change a Gram matrix. If a device took $|\psi\rangle|0\rangle$ to $|\psi\rangle|\psi\rangle$ for two inputs with overlap x, the Gram matrix of the outputs, with entries x^2, would have to equal that of the inputs, with entries x. With an environment the requirement becomes $x = x^2\langle E_\psi \mid E_\phi\rangle$, which still forces $x \in \{0,1\}$ (literature check, derivation) [V-c]. REG c5 wrote the condition as a positivity test, $X^{(1)} - pX^{(2)} \succcurlyeq 0$, and read it at two points. At p = 1 the matrix is $\begin{pmatrix} 0 & x - x^2 \\ x - x^2 & 0 \end{pmatrix}$, with eigenvalues $\pm x(x-1)$. It is admissible only at x = 0 and x = 1, a result REG c5 prints as "published as NO-CLONING" [V]. Check 9 confirms the two-point set symbolically, on 100,001 real overlaps, and on a 201 × 360 polar grid of complex overlaps [V].

The result belongs to Wootters and Zurek (**Nature 299, 802 (1982)**) and to Dieks (**Phys. Lett. A 92, 271 (1982)**). Park's 1970 paper anticipated it, as Ortigoso's history documents (**Park 1970**; **Ortigoso 2018**). Bibliographic records only; the texts were not read [V-c].

7.2 Duan–Guo is the boundary of the same test, with one correction

Lowering p below 1 opens the admissible set. For two states the smallest eigenvalue of $X^{(1)} - pX^{(2)}$ is $(1-x)\big(1-p(1+x)\big)$ whenever $p \le 1/x$ (algebra checked in this research) [V-c], so the boundary is

$$p_{\max} = \frac{1}{1+x}.$$

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The programme found it three ways without being told the answer: by root-finding on the eigenvalue (SIG c5, 7 of 7), by building an explicit one-Kraus cloner with the largest p its norm allows (ARROW c7, 7 of 7 to twelve digits), and by **gram_gap** (REG c5) **[V]**. Check 7 extends the comparison to 1000 overlaps in [0, 0.999], with a maximum deviation of 6.55×10^{-1} and a sharp feasibility edge **[V]**. Duan and Guo's full text was read. Theorem 1 says states are probabilistically clonable if and only if they are linearly independent. Theorem 2 gives the matrix condition, and for two states $(\gamma_1 + \gamma_2)/2 \leq 1/(1 + |\langle \Psi_1 \mid \Psi_2 \rangle|)$ (**Duan & Guo, arXiv:quant-ph/9804064; PRL 80, 4999 (1998)**) **[V]**.

Reading the paper forced a correction. Duan and Guo's matrix is $X^{(1)} - \sqrt{\Gamma}\, X_P \sqrt{\Gamma}^{\dagger}$, where X_P carries the overlaps of the device's probe states. The registry's bare form, with $X_{ij}^{(N)} = \langle \psi_i \mid \psi_j \rangle^N$, is the special case of identical probes. It is exact for two states. For three or more states it is sufficient but not necessary: in 2149 of 4000 random linearly independent three-state sets, orthogonal probes beat identical ones, the largest gap being 0.374 against 0.496 (literature check, numerical) **[V-c]**. The exact criterion asks for some probe Gram matrix P, positive semidefinite with unit diagonal, such that $X^{(1)} - \sqrt{\Gamma}\left(X^{(N)} \circ P\right)\sqrt{\Gamma}^{\dagger} \succcurlyeq 0$ **[V-c]**. There is also a genuine discontinuity at x = 1. Identical states can be "copied" with p = 1, while $1/(1 + x) \to 1/2$ as $x \to 1^{-}$, so the named rate holds on [0, 1) and jumps at the edge (Check 9) **[V]**. The two-state bound generalizes to M → N copies as $(1 - s^M)/(1 - s^N)$, attributed to Chefles and Barnett (**J. Phys. A 31, 10097 (1998)**); that attribution was not verified in their text **[V-c]**.

7.3 Bužek–Hillery and Massar–Popescu are points on one surface

Universal cloning, the kind that must work for every input state, has its own boundary: the Gisin–Massar fidelity of an optimal N → M qubit cloner,

$$F(N, M) = \frac{M(N + 1) + N}{M(N + 2)}.$$

Two named results are points on this surface. Bužek and Hillery's 5/6 is F(1, 2). Massar and Popescu's optimal estimation fidelity $(N + 1)/(N + 2)$ is its $M \to \infty$ edge, the "cloning ↔ estimation" link that Bruß, Ekert and Macchiavello named, and Werner's formula extends it to d dimensions (**Bužek & Hillery 1996**; **Gisin & Massar 1997**; **Bruß, Ekert & Macchiavello 1998**; **Werner 1998**; **Massar & Popescu 1995**) **[V-c: records confirmed, texts not read]**. Check 7 verifies the algebra exactly: F(1, 2) = 5/6, F(N, N) = 1 for N up to 200, limit $(N + 1)/(N + 2)$, and gap $N/\big(M(N + 2)\big)$. An explicit Werner cloner reaches F for 26 (N, M) pairs to 6.7×10^{-16} **[V]**. That is attainment, not optimality; the optimality half rests on the literature **[V-c]**.

The programme reached the same numbers by optimization. A symmetric 1 → 2 cloner scored on both output marginals over the six-state octahedron design reached 0.83333333 at every ancilla dimension from 2 to 4, and 0.81903559 at $d_a = 1$; the identity map scores 0.75 (CEIL c9) **[V-c: optimizer, not proof]**. Monte Carlo reproduced $(N + 1)/(N + 2)$ for N = 1 to 10 to within 2.3×10^{-4}. At N = 50 it drifted by 5.05×10^{-3}, diagnosed as a uniform sampler undersampling a concentrated peak and never re-run (SIG c9) **[V-c]**. REG's statement that "Optimizing over the same admissible set (a variational step, not just positivity) gives 5/6 … and (N+1)/(N+2)" is true of the literature. In the programme, however, neither value was emitted from the Gram condition itself **[O]**. Figure 4 draws both panels and marks the difference.

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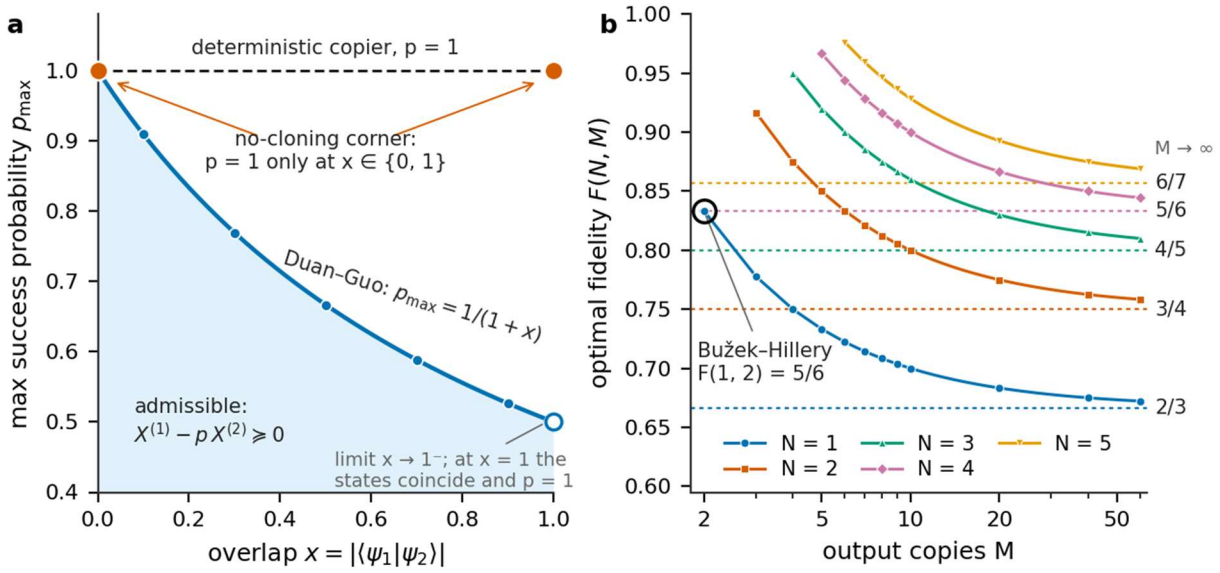


Figure 4. *Copying as a Gram-positivity boundary. (a) For two pure states with overlap x, exact cloning with success probability p is admissible if and only if $X^{(1)} - pX^{(2)} \succeq 0$. The frontier is Duan–Guo's p_max = 1/(1+x); the deterministic copier (p = 1) is admissible only at x ∈ {0, 1}, which is the no-cloning theorem; points are computed from the inequality. At x = 1 the states coincide and p = 1, so the rate jumps at the edge. (b) Optimal universal N → M qubit-cloning fidelity F(N, M) = [M(N+1) + N]/[M(N+2)] (Gisin–Massar), with Bužek–Hillery's F(1, 2) = 5/6 circled and the M → ∞ limits (N+1)/(N+2), the optimal estimation fidelities of Massar–Popescu. Panel (b) plots the closed form, not an emission from the Gram inequality.*

7.4 Laplace's rule sits at the edge of the surface

The edge value $(N+1)/(N+2)$ is Laplace's rule of succession, and the identity is exact rather than an analogy. For the covariant measurement on N copies, the optimal fidelity is $F = E[p^{N+1}]/E[p^N]$ with $p = |\langle\psi \mid \phi\rangle|^2$: the predictive probability that one more trial succeeds. For qubits, p is uniform on [0, 1] under the Haar measure, so

$$F = \frac{\int_0^1 t^{N+1}\, dt}{\int_0^1 t^N\, dt} = \frac{N+1}{N+2},$$

which is Laplace's rule after N successes in N trials with a uniform prior (**Rule of succession**) [V-c]. In d dimensions the overlap is Beta(1, d − 1) and the fidelity is $(N+1)/(N+d) = \dim\mathrm{Sym}^N/\dim\mathrm{Sym}^{N+1}$, the d-category rule of succession with a uniform Dirichlet prior (literature check, symbolic) [V-c]. Check 7 evaluates the Laplace integral, the Beta(N+1, 1) mean and the estimation integral side by side for N from 0 to 200, and they agree to 1.9×10^{-15} [V]. The code itself notes that two of those integrands are the same polynomial. The non-trivial inputs are the uniform law of the qubit overlap, which a 200,000-sample Haar check supports (mean 0.50004, variance 0.08321 against 1/12, KS p = 0.939, outside the PASS condition), and the covariant-measurement fidelity, which is taken from the literature [V-c]. No published source stating the Laplace link was found. Candidates to check before any novelty claim include Bruß and Macchiavello (1999), Harrow's "church of the symmetric subspace", Chiribella's work on cloning and de Finetti theorems, and Blume-Kohout (2010) [O]. The derivation is the paper's; its novelty is unestablished.

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7.5 Planck's address: Robertson–Schrödinger is Gram positivity, and Law 2 reads as it

REG's opening claim named Planck among the libraries, "Planck, Gödel, Nyquist, Barkhausen, Landauer and the rest", but emitted nothing for him (REG co) **[V]**. The placement this paper adds puts the Planck/Heisenberg–Robertson floor at the same root as no-cloning.

Take any two observables A and B and a state ψ, and form the deviation vectors $u = (A - \langle A\rangle)\psi$ and $v = (B - \langle B\rangle)\psi$. Their Gram matrix is

$$G = \begin{pmatrix} \langle u \mid u\rangle & \langle u \mid v\rangle \\ \langle v \mid u\rangle & \langle v \mid v\rangle \end{pmatrix}, \qquad \langle u \mid v\rangle = [1/2\,\langle\{A,B\}\rangle - \langle A\rangle\langle B\rangle] + i\,\frac{1}{2i}\langle[A,B]\rangle.$$

A 2 × 2 Hermitian matrix with non-negative diagonal is positive semidefinite exactly when its determinant is non-negative, and $\det G \geq 0$ is the Robertson–Schrödinger relation

$$\sigma_A^2\sigma_B^2 \;\geq\; (1/2\,\langle\{A,B\}\rangle - \langle A\rangle\langle B\rangle)^2 + (1/2i\,\langle[A,B]\rangle)^2.$$

Gram positivity is therefore equivalent to Robertson–Schrödinger, not merely sufficient for it. Robertson's $\sigma_A\sigma_B \geq 1/2\,|\langle[A,B]\rangle|$ is the weaker statement that drops the covariance, and equality holds exactly when u and v are parallel (literature check, derivation with 20,000 random tests) **[V-c]**. Check 5 ran 10,000 fresh random triples (A, B, ψ) in dimensions 2 to 8. There were zero violations of $\det G \geq 0$, of Gram positivity, of Robertson and of Robertson–Schrödinger at relative tolerance 10^{-12}, and the identity $\det G$ = RS slack held to 3.7×10^{-12} **[V]**. Because a Gram matrix is always positive semidefinite, Part A can fail only by rounding. The informative content is the identity, and the identity is what places the uncertainty relation at root 2 **[V]**.

The constant enters through translations. The translation operator $T(a) = e^{-iap/\hbar}$ satisfies $T(a)^\dagger x\, T(a) = x + a$, and differentiating at a = 0 gives $[x,p] = i\hbar$. So "momentum generates translations" and the canonical commutation relation are one statement read infinitesimally, and substituting $\langle[x,p]\rangle = i\hbar$ into Robertson gives Kennard's $\sigma_x\sigma_p \geq \hbar/2$ (literature check, derivation) **[V-c]**. On a 16,384-point grid, Gaussian packets sit exactly on the floor ($\Delta x\,\Delta p$ = 0.5000000000, with ħ = 1, including a displaced and boosted packet). A chirped Gaussian sits above Robertson at 0.7810249676 but saturates Robertson–Schrödinger. Non-Gaussian packets lie above both: 0.9264353650 for a smoothed square pulse, 1.4934917497 for a two-Gaussian superposition, and exactly 1.5 for the first excited state. $\mathrm{Im}\langle xp\rangle$ reads 1/2 in every case (Check 5) **[V]**. Figure 5 shows the floor across a family of shapes. Only the Gaussian touches it; the two-sided exponential packet gives $1/\sqrt{2}$; cusps and boxes diverge **[V]**.

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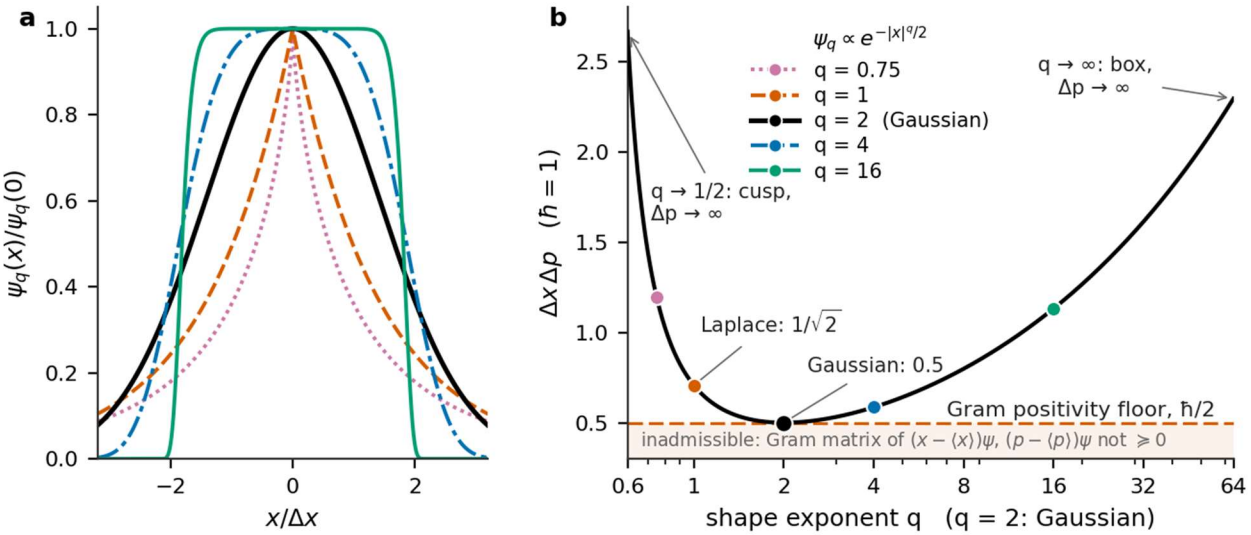


Figure 5. *The uncertainty floor as Gram positivity. (a) Real wavepackets ψ_q ∝exp(−|x|^q/2), from cusped (q < 2) through Gaussian (q = 2) to flat-topped (q > 2), scaled to unit Δx. (b) Their product ΔxΔp (ħ = 1) from the closed form, checked by quadrature and FFT. It touches the Robertson floor ħ/2, which is positivity of the Gram matrix of (x − ⟨x⟩)ψ and (p − ⟨p⟩)ψ, only at the Gaussian. It equals 1/√2 for the two-sided exponential packet and diverges toward the cusp (q → 1/2) and box (q → ∞) limits.*

The author's Law 2 states: "no matter may hold a location without the potential to be moved from that location." Read against the mathematics above, the potential to be moved is momentum, the generator of translations, and holding a location means shrinking σ_x. The Gram inequality then says the hold has a price that diverges. The constraint cascade executed exactly this before the placement was made. Its rung 2, headed "LAW 2: NO LOCATION MAY BE HELD. THE HOLD HAS A DIVERGING PRICE", found a discretized ground-state energy of 0.499999319221, $\langle p^2\rangle$ = 0.500000000 ("NOT ZERO. it moves") and $\sigma_x\sigma_p$ = 0.4999993192. Its "hold tax" $\min\langle p^2\rangle/2 = 1/(8\sigma_x^2)$ is 12.5 at $\sigma_x = 0.1$, 1.25×10^5 at 10^{-3} and 1.25×10^{11} at 10^{-6}. It concluded: "A held location is not forbidden by fiat; it is priced out of the universe." (MN L85107–85128) **[V]**. The address of Planck's floor is (Gram positivity | a pair of observables whose commutator is a multiple of the identity | ħ | the product $\sigma_x\sigma_p$), and on this reading Law 2 is the uncertainty relation stated as a constraint on location **[H]**.

The placement has limits, and they are standard. Robertson with $[x,p] = i\hbar$ needs ψ in the domains of xp and px, and naive use fails for angle and angular momentum or for a particle on an interval. The canonical relation has no finite-dimensional representation (take the trace of both sides), so Check 5's random matrices test Robertson for general A and B, not the canonical relation itself, and its grid reading of 1/2 relies on packets sitting well inside the box **[V-c]**. The Cauchy–Schwarz derivation is textbook material; its standard homes in Griffiths, Sakurai, and Nielsen and Chuang are recalled, not read. No verified source was found that names the 2 × 2 case a Gram matrix, although Robertson's 1934 determinant relation for several observables is the n × n Gram determinant (**Robertson 1929**; **Robertson 1934**; **Kennard 1927**; **Heisenberg 1927**; Schrödinger 1930 in the **Angelow–Batoni translation**) **[V-c]**. What is new is the address, not the inequality. No-cloning, Duan–Guo and the uncertainty floor all live where one Gram matrix stops being positive.

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The same move suggests renaming the root. REG called root 2 "inner-product preservation", which describes how the Copy members arise: isometries preserve Gram matrices. The condition that actually emits no-cloning, Duan–Guo and Robertson–Schrödinger is positivity of a Gram matrix. This paper therefore names the root **Gram positivity** and treats inner-product preservation as its Copy reading [S].

8. Roots 3 and 4: dilation and counting measure, grouped by assertion

Two further roots complete the registry, and the registry notebook is candid about their standing: "Roots 1 and 2 are verified above — each emits its members from a single stated condition. Roots 3 and 4 are asserted at this level of grouping; I have checked members of 3 individually but not that the grouping is minimal." (REG c6) [V as a statement of status].

Root 3, dilation or set completion, is a dimension count. Its members are three. The first is that the minimal environment of a channel equals its Choi rank [V-c, six frontier points]. The second is ancilla saturation: the cloning frontier stops moving at $d_a = 2$ [V]. The third is the strict inclusion of trace-preserving maps in trace-non-increasing ones, TP ⊊ TNI, which is what heralding buys [V]. The literature supplies the theorem behind the first member, and the literature check verified it with a three-line proof plus numerics [V-c]. Write the Choi matrix as $J(\Phi) = \sum_i |K_i\rangle\rangle\langle\langle K_i|$, a sum over vectorized Kraus operators. Then rank J cannot exceed the number of Kraus operators, and the spectral decomposition of J attains it. Every Stinespring isometry V yields Kraus operators $(I \otimes \langle i|)V$, so the environment needs at least rank J dimensions, and the minimal Kraus set attains that too. Random 3 × 3 channels built from k = 1, 2, 4 and 9 Kraus operators returned rank k, with an isometric dilation of exactly that dimension. The same check gives $\sum_i K_i^\dagger K_i = I$ exactly when $\mathrm{Tr}_{\mathrm{out}} J = I$, and $\leq I$ for the trace-non-increasing case (Stinespring 1955; Choi 1975; Kraus 1983).

Root 4, counting measure on a finite substrate, is a measure. It has two members. One is the noise law $\mathrm{CV} \sim \Omega^{-1/2}$ of Section 5 [V]. The other is Landauer's $kT\ln 2$, which has never been computed anywhere in the corpus [O]. The Erase verb appears only in prose. VRT c17 says the hysteresis area of a switching latch "is Landauer's kT ln2 in chemical clothing"; a plan to compute that area was never run; and CEIL predicted that Erase would be "a price that goes to zero with temperature — and it does", stated rather than computed [O]. At 300 K, $k_B T\ln 2 = 2.871 \times 10^{-21}$ J = 17.92 meV (literature check) [V-c]. The counting reading of the principle is standard: erasing a random bit halves the number of logical states, so an entropy of $k_B \ln 2$ must go to the environment (Landauer 1961; Bennett 2003; Bérut et al. 2012) [V-c]. Bérut's page range is given as 187–189 in one record and 187–190 in another.

REG argued that the roots do not reduce to one another: "Root 1 is a spectral condition on a transfer product. Root 2 is a positivity condition on a Gram matrix. Root 3 is a dimension count. Root 4 is a measure. Nothing in root 1 produces an inner product, and nothing in root 2 produces a frequency." (REG c6) [V as argument]. It also named its own falsifier: "a demonstration that two of the four roots do reduce to one. I have not tried hard to collapse roots 3 and 4 into the others, and that is the obvious next check before trusting the number four." (REG c8) [O].

This paper's placements bear on that check without settling it. The Choi matrix is positive semidefinite, $J = KK^\dagger$ with the vectorized Kraus operators as the columns of K, and it has the same rank as their Gram matrix $K^\dagger K$. Root 3's dimension count is therefore the rank of a root-2 object [V-c]. The Laplace link makes

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$(N+1)/(N+2)$ a ratio of symmetric-subspace dimensions, $\dim\mathrm{Sym}^N/\dim\mathrm{Sym}^{N+1}$, so the same value has an address at root 2 (a point on the cloning surface) and at root 4 (a ratio of counts) **[V-c]**. And $kT\ln 2$ is the logarithm of a count. One reading is that roots 3 and 4 are the rank and the trace-and-dimension readers of positive operators, that is, readings of root 2 rather than independent roots. That reading would cut the count from four to two, loop closure and Gram positivity **[H]**. It is precisely the collapse REG asked for, and it has not been tested **[O]**.

9. The registry and the runtime it lives in

9.1 Eighteen packages, four roots, one empty slot

Figure 6 draws the registry as a periodic table. Eighteen named packages from the loop/channel corpus fall under four roots: nine under loop closure, four under Gram positivity, three under dilation and two under counting measure (REG c6) **[V as a count]**. The figure script parses REG's table and asserts that its tiles are exactly REG's packages. It shows the resonant rule as the exact $2/Q + 1/Q^2$ that REG emits, where REG's table writes "2/Q" **[V]**. Two marks record the corpus's own qualifications. A dagger marks 5/6 and $(N+1)/(N+2)$, which the programme reached by optimization and Monte Carlo rather than from the Gram condition. A double dagger marks $kT\ln 2$, which was never computed. The fifth column is empty.

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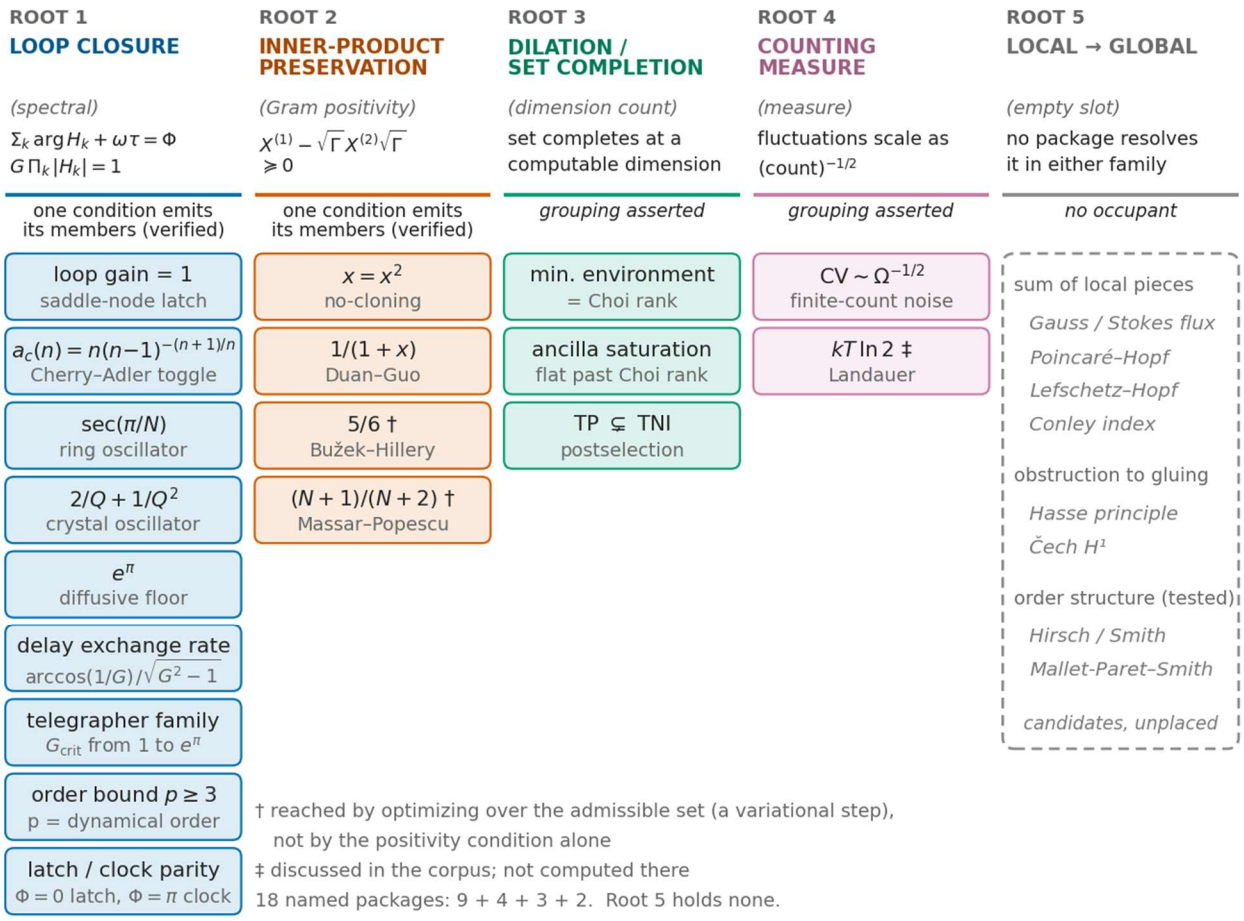


Figure 6. The registry as a periodic table. Eighteen named packages from the loop/channel corpus fall under four roots of different mathematical kind: loop closure (spectral, 9 packages), inner-product preservation (Gram positivity, 4), dilation/set completion (dimension count, 3) and counting measure (2). For roots 1–2 one stated condition emits every member (verified; † marks the two members reached by optimizing over the admissible set rather than by positivity alone). For roots 3–4 the grouping is asserted, and ‡ marks kT ln 2, which is discussed but not computed. The fifth slot, local → global, is empty in both families of the corpus; the grey entries are candidate occupants of three kinds: sums of local pieces, obstructions to gluing, and order structure, the one kind that has passed a test (Section 10.1).

REG's verdict on its own claim is sharp:

> *The registry is real. It has four roots, not one. (REG c6)*

The claim it tested was that "Planck, Gödel, Nyquist, Barkhausen, Landauer and the rest are not separate discoveries but libraries in one registry", and REG c8 says of it: "It is wrong about cardinality." **[V]** This paper extends the registry without changing its shape **[H]**. Griffith's 8 joins root 1 at the three-lag address (Section 4.5). Planck's floor and Laplace's rule join root 2 (Sections 7.4–7.5). The inverse-square flux law becomes the first package placed in the empty slot's first mechanism (Section 10.3), and order structure the first candidate there to pass a test (Section 10.1). Lawvere's diagonal becomes a candidate root of its own (Section 11). The conserved-mass threshold M_c is still an address without a resolver.

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9.2 The runtime as software documentation

The author's instruction of 26 September 2026 was to treat the work "as a software documentation", on the principle that math laws are libraries reality can pull from. The resulting NEXUS Runtime Specification v0.1 is a DRAFT CORE SPECIFICATION with an explicit non-goal: it "does not assert that every mathematical or physical law has been derived from NEXUS. It specifies how existing laws can be registered, invoked, validated, hidden, refactored, or extended as transformation libraries." [S] Its central statement is "NEXUS defines the runtime grammar; mathematical and physical laws are libraries that implement lawful transformations within it." [S] Its runtime loop runs STATE → INTERFACE/CUT → READER QUERY → ADMISSIBILITY CHECK → LIBRARY RESOLUTION → TRANSFORMATION EXECUTION → INVARIANT/LEDGER UPDATE → NEW STATE → RECONDITIONED INTERFACE, and around again [S]. Among its eight core objects, a Reader is "a bounded system that retains a quotient of history sufficient for future continuation", and a Library is "a reusable family of constraint-preserving transformations", with the boxed gloss "Law = reusable constraint-preserving transformation specification" [S].

The address scheme is the part of that grammar that has been executed. Table 5 maps each slot of the address onto the specification's objects [H as a mapping; S for the objects].

Table 5. The four address slots mapped onto the objects of the runtime specification.

Address slot	Runtime object	Contract method
root ρ	library namespace (the admissibility condition a library family implements)	**describe(), validate()**
class or signature Σ	transformation class $\mathcal{T}_\Sigma$	**signature(), preconditions()**
coordinate θ	parameters, resource coordinates	**transform(state, parameters), cost()**
reader R	reader–interface bond	**expose(), hide(), identify()**
resolver	LIBRARY RESOLUTION step	**preconditions()** then **transform()**
the boundary itself	failure mode at the edge of admissibility	**fail()**

The mapping is closer than an analogy. The specification's **preconditions()** must distinguish four ways a call can fail: "unavailable method; available method with insufficient resource; method absent from current transformation class; method excluded by the current interface" [S]. The last three are the three motions of Section 2.2. Insufficient resource is a coordinate that has not moved far enough; absent from the class calls for an arrow; excluded by the interface is a reader change. The specification's two-level rule states it outright: "Class extension opens a coordinate. Coordinate values are then spent inside the opened class." [S]

The five meta-operations are the specification's other load-bearing element, and unlike the rest of the runtime they have executed backing [V]. The backing comes from the token-arena programme of the carry/reader/cohomology family, which measured what happens to a space of residual transformations when the substrate is edited, symmetrized, subdivided, merged or pruned. Table 6 pairs each operation with its evidence.

Table 6. The five runtime meta-operations and their executed backing in the arena programme.

Meta-operation	Specification	Executed backing (arena §10–§12)	Status
EXTEND	"changes what transformations exist for the reader", $\mathcal{T}_\Sigma \subsetneq \mathcal{T}_{E(\Sigma)}$	edge edits: map onto, cokernel 0 in 6 of 6 cases (e.g. $P_6 \to C_6$: 30 transformations survive, 6 new, 0 destroyed)	[V] measured, not proved

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Meta-operation	Specification	Executed backing (arena §10–§12)	Status
HIDE	"hidden ≠ destroyed"	symmetry steps: $R_{G'} \hookrightarrow R_G$ injective, kernel 0 (C_6, raising the symmetry from Z_2 to D_6: dim R from 7 to 0, kernel 0, cokernel 7)	[V]
REFACTOR	"the transformation persists while its representation changes"	subdivision: chain-map identity $\partial' F = F\,\partial$ to 0.0 exactly in all five cases	[V]; later proved
IDENTIFY	"the quotient operation that creates runtime-visible objects"	merging: q_R^* injective, kernel 0	[V] proved
EXCLUDE	"the only operation that actually removes admissible continuation"	collision: kernel equals the Euler count of the excluded star; kernels 3, 4, 5, 6, 10, 4, 1, 3 across eight contractions	[V]

The arena's boxed summary is "edits extend or re-factor; symmetry hides; merging identifies; only exclusion erases", together with "The lossy half is not the many-to-one merge. It is the prohibition on two occupancies sharing one location." (MN L109030–109042) [V]. The runtime's rule that change is not loss therefore has a measured basis, and so does its location clause: the one operation that destroys structure is two occupancies forced into one place. The phrase "hidden ≠ destroyed" is the specification's wording. The notebooks say "R_{D_6} = 0 is a statement about the observer, not about the arena" and "projection erases visibility, not structure" [V].

Above everything sits a dependency graph. It runs from C0 to DISTINCTION to C1/CONTINUATION, and C1's children are RETAIN, ACCUMULATE, COMMIT, FOLD, INTERFACE/CUT (with IDENTIFY, HIDE and EXPOSE), READER (quotient state, access geometry, method resolution), TRANSFORMATION LIBRARIES, CLASS EXTENSION ("opens new coordinates"), RESOURCE COORDINATES, and NEW STATE, which loops back to a reconditioned interface [S].

The specification also carries honest limits, and this paper keeps them.

- It is an [S] artifact written by an assistant, and nothing in the corpus implements the **LibraryContract** except the loop-closure resolver [S].
- There are two incompatible API sketches: the twelve-method **LibraryContract**, and a thirteen-method **LawLibrary** that adds **classify_extension()**, **declare_access_geometry()** and **scope()** [S].
- Of the eight status markers it requires, only four are defined in prose [S].
- Its "Current Work Registry" lists nine loop/channel modules and omits the arena family entirely, although that family holds the only executed evidence for the meta-operations [R as an omission].
- Its own final caution is the right one: "This is a library architecture, not an ontology claim that all branches are literally the same equation." [S]

10. The empty slot: local → global

10.1 A local condition that is right about a state and wrong about the system

REG's Step 5 looked for a fifth root and found a hole instead. The test object was a delayed four-stage positive-feedback Hill ring at gain G = 5. The local package gives a precise prediction: the symmetric state loses stability through a Φ = 2π Hopf pair once $\tau > \left(2\pi - 4\arccos(1/G)\right)/\sqrt{G^2-1} = 0.1644$ (REG c7) [V]. Twelve random constant histories were integrated at each delay. The cycle was found in 0, 0, 1, 1, 2, 5 and 9 of 12 runs at τ = 0.5, 1.0, 1.5, 1.75, 2.0, 3.0 and 5.0. REG read this as a basin fraction running 0 → 0.083 →

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0.167 → 0.417 → 0.750, and "The cycle first appears between tau = 1.0 and 1.5. A factor of about eight, in the direction of the package being too permissive." **[V]** The reading: the local condition "correctly locates a Hopf bifurcation of the symmetric state. But that state is already unstable at G>1 … The cycle's actual birth is a global event", and "Existence of the cycle is a threshold; its basin is a continuous coordinate" (REG c8) **[V]**.

The onset depends on sampling, and three runs bracket it. An earlier independent run with 8 histories, a different initial box and a longer horizon first saw the cycle at $\tau = 2.0$ **[V-c]**. A dense re-run for this paper copied REG's model, integrator and cycle criterion and used 96 histories at each of 17 delays. It reproduced REG's twelve-history numbers exactly under REG's seed, and it placed the first cycle at $1.25 < \tau \leq 1.5$, which is 7.6 to 9.1 times the local value (fig7_rerun) **[V]**. The three brackets together put the first oscillation still present at the run length at roughly 6 to 12 times the local value **[V]**. The dense curve rises through 0.031, 0.115, 0.188, 0.250, 0.208, 0.385, 0.615, 0.854, 0.781 and 0.896 at $\tau = 1.5, 1.75, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0$ and 6.0 **[V]**. It agrees with REG's values within binomial error everywhere except $\tau = 3.0$. There REG's 5/12 = 0.417 has a 95% interval of [0.193, 0.680], while the dense estimate is 20/96 = 0.208 with interval [0.139, 0.300]. The occupancy curve is therefore "rising, noisily" rather than the clean ladder REG printed **[R as wording]**. "First cycle detected" is a detection bound: at $\tau = 1.25$, zero of 96 still allows an occupancy of up to about 3.8% **[V-c]**. Figure 7a shows all three runs against the local prediction.

The horizon test withdraws the cycle. A review of this paper in a separate session pointed out what the registry had missed. The ring's four links are all repressive, so its sign product is positive, and a positive cyclic loop with delay is a monotone system. For strongly monotone systems, Hirsch's theory and its extension to delay equations make generic orbits converge to equilibria and rule out attracting nontrivial periodic orbits, while positive delayed feedback is known instead for oscillating transients that live longer as the delay grows (Hirsch 1988; Smith & Thieme 1991; Smith 1995; Pakdaman, Grotta-Ragazzo & Malta 1998) **[V-c: theorems cited, not re-derived here]**. The review proposed a cheap decisive test: hold τ fixed and lengthen the run. It was run on REG's exact model, design, integrator, initial histories and cycle criterion, continued to eight times REG's run length, with 48 histories whose first 12 are REG's (**horizon_test.py**). REG's counts are reproduced exactly at T = 500: 1, 2, 5 and 9 of 12 at $\tau = 1.5, 2.0, 3.0$ and 5.0. Then, at every delay, the count falls as the run grows (Figure 7b) **[V]**.

Horizon test on the registry's delayed ring: histories (of 48) still oscillating at run length T, at fixed delay τ.

τ	T = 500	T = 1000	T = 2000	T = 4000
1.5	3	0	0	0
2.0	9	5	0	0
3.0	17	14	13	10
5.0	37	34	27	20

The histories still oscillating at T = 4000 swing at full amplitude (median swing 0.997) until they drop onto a latch, which is how a long transient ends, not how a cycle decays **[V-c]**. REG's twelve histories fall from 1, 2, 5 and 9 to 0, 0, 3 and 5 **[V]**. REG's "basin fraction" was therefore a finite-horizon oscillatory occupancy, Occ(τ, T), and its rise with τ at T = 500 measured transient lifetime, not an attractor **[R]**. This is error E8, the slow transient read as a cycle, for the fourth time (Section 15). The programme's own rule would have caught it: run length must scale with the slowest relevant time, and near a long-lived transient that time is the transient's lifetime.

What survives is sharper. Three readers looked at one system and returned three verdicts. The local spectrum of the symmetric state says a periodic orbit is born at $\tau > 0.1644$; REG was right that this correctly

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locates a Hopf bifurcation of a state that is already unstable [V]. The finite-horizon simulation says an oscillating occupancy rises with τ [V]. The sign pattern of the interaction graph says that no stable cycle exists at any τ, and the horizon test agrees [V-c]. The system never changed. What changed was each reader's access, and the simulation reader carried a hidden argument of its own, the run length T: the quantity it reported was Occ(τ, T) with T silently fixed [H]. That is the Sun() thesis applied to an observer, and it is the most useful thing the empty slot has produced so far.

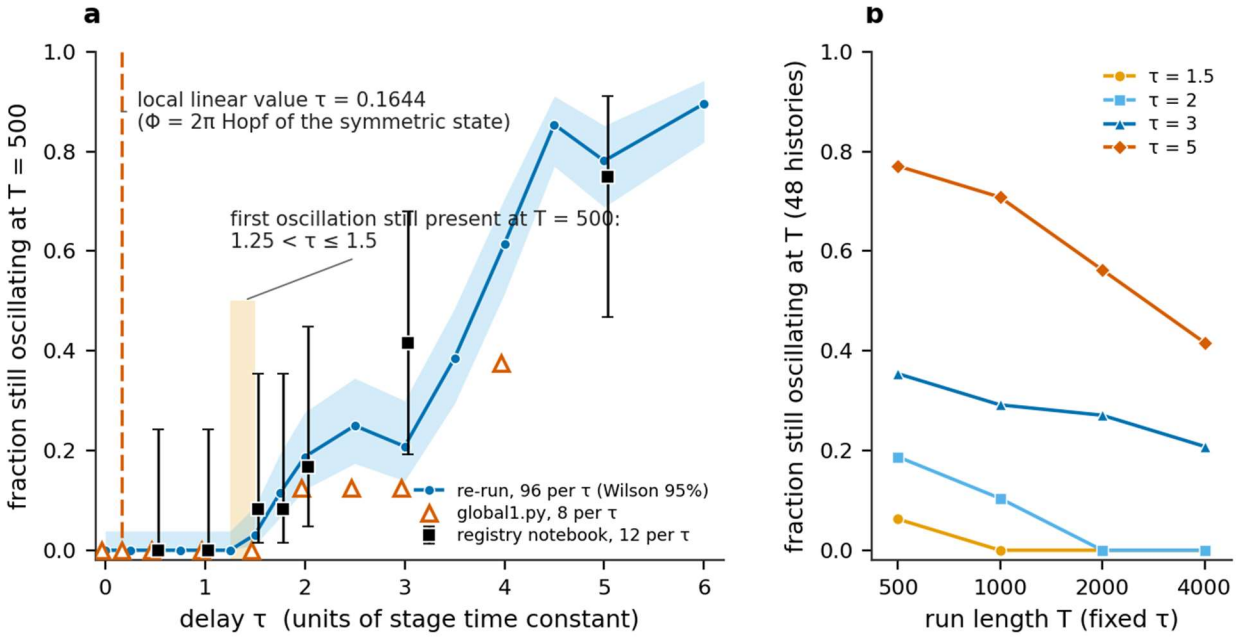


Figure 7. A local condition, a finite run and the sign pattern disagree about the delayed four-stage positive-feedback ring (G = 5). (a) The local Φ = 2π Hopf condition of the symmetric state predicts oscillation for τ > 0.1644 (dashed line). At a fixed run length T = 500, oscillation is first still present from 1.25 < τ ≤ 1.5, and the fraction of random constant initial histories still oscillating at T rises with τ. Blue: 96 histories per τ with Wilson 95% band. Black squares: the 12-history values reported in the registry notebook, reproduced exactly. Triangles: an independent 8-history run with a different initial box and horizon. Markers are offset by ±0.035 in τ. (b) Holding τ fixed and lengthening the run, the same fraction falls at every delay (48 histories, the first 12 being the registry's). The oscillations are long transients, as monotone-systems theory requires of a positive loop, so panel (a) measures transient lifetime, not a basin.

10.2 The same gap in the other family

REG then named three results from the carry/reader/cohomology family as the same gap seen on another machine (REG c8) **[V as a citation; H as a reading]**.

The first is §13, "The capacity defect is a Betti number — and the ladder does not always terminate". It showed that the defect when capacity rises by one equals $\beta_1(Q_r) = |E(Q_r)| - |V(Q_r)| + c(Q_r)$ of a cone quotient graph, 55 of 55 with zero mismatches. It also showed that a terminal zero in the ladder "was a property of the *local symmetry at the distinguished site*, not of capacity saturation": a prism at K = 4 and a path with an interior site both break it (MN L108201–108554) [V].

The second is §9, "The reader bridge — an explicit map, and what it misses". It showed that every reader cochain is a pullback from the base graph, found that symmetric bases reach nothing, and found that trivial-automorphism bases reach exactly $\varepsilon(X, 1; G)$ in 19 of 19 cases. It concluded: "So the reader does not lose the residual — it never had access to it" [V]. What no single-token observable can reach, $\varepsilon(X, K) - \varepsilon(X, 1)$, is

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loop structure created by putting K tokens on X. C_8 at K = 4 has three such channels, and a reader can reach none of them **[V]**.

The third is $\varepsilon(X, K; G) = \beta_1(A /\!\!/ G)$, an edge-orbit residual that a local derivative cannot see **[V]**.

REG's conclusion: "local accessible structure ⊊ global relational structure", and "the missing dependency is identical … not because their answers match, but because their gaps match." **[O as a stated conclusion]** Its boxed line reads:

> *Four roots, and a fifth slot that is empty in both families: local → global. (REG c8)*

The two families share no computation. The mirror is a match of headings and structure, not a calculation connecting a delayed ring's dynamics to a Betti number **[H]**. On the loop side, the horizon test changed what the mirror shows. The gap between local certificate and global behaviour was real, but it was not empty of theory: for a sign-structured loop the global package is order structure (Section 10.3). Whether the carry family's gap closes the same way, by an order-preservation argument rather than a Betti number, is the forward question. If it does, the two families share a root and not only a gap **[O]**.

10.3 Two mechanisms for one slot, and where the inverse-square law lives

The literature check found that "local → global" names two different mechanisms, and a registry should give them separate addresses **[V-c]**.

The first is the **sum of local pieces**: a global invariant equals a sum or integral of local data, with no obstruction term. Gauss's divergence theorem and Stokes's theorem are the parents. Poincaré–Hopf makes the sum of vector-field indices equal the Euler characteristic, and Lefschetz–Hopf makes the sum of fixed-point indices equal the Lefschetz number. Morse theory bounds Betti numbers by counts of critical points. The Conley index extends the family to flows through the Ważewski principle: "If the index is nontrivial then the invariant set S is nonempty" (**Lefschetz**; **Poincaré–Hopf**; **Morse theory**; **Conley index**) **[V]**.

The second is the **obstruction to gluing**: compatible local solutions can fail to come from a global one, and the failure is itself a cohomology class. The Hasse principle is the classic case. Selmer's cubic $3x^3 + 4y^3 + 5z^3 = 0$ has real and p-adic solutions but no nontrivial rational one, and the obstruction is Brauer–Manin (**Hasse principle**) **[V]**. In a modern information-theoretic setting, Abramsky, Mansfield and Barbosa define a Čech cohomology class that "vanishes if the family has a global section" (**Abramsky et al. 2012**) **[V]**.

A third mechanism, which the review supplied and the horizon test confirmed for the loop family, is **order structure**. When the signs of an interaction graph make a system monotone, global conclusions follow from the sign pattern alone, and no Jacobian at a point can supply that pattern. Hirsch's generic-convergence theorem, extended to delay equations by Smith and Thieme, makes almost every orbit of a strongly monotone system converge to an equilibrium and forbids attracting nontrivial periodic orbits; Mallet-Paret and Smith's Poincaré–Bendixson theorem constrains cyclic feedback of either sign **[V-c: cited, not re-derived]**. For the delayed positive ring this settles the question the local package could not **[V-c]**.

The two families appear to need different mechanisms **[H]**. The loop family's gap is an existence question about invariant sets, and its natural home is a Morse–Conley decomposition of the global attractor into local attractors, repellers and connecting orbits. For positive loops that gap is closed by order structure; for negative loops and other global questions, existing theory constrains but does not close it. Mallet-Paret and Sell prove a Poincaré–Bendixson theorem for monotone cyclic feedback with delay, which classifies each ω-limit set but says nothing against several coexisting. Krisztin and Vas construct large-amplitude periodic

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orbits coexisting with stable equilibria in delayed positive feedback (**Mallet-Paret & Smith 1990**; **Mallet-Paret & Sell 1996**; **Krisztin & Vas 2011**) **[V-c]**. The carry/reader family's gap is a first Betti number that pulled-back readers cannot see, which reads like the second mechanism, although it is computed as an Euler-type count of the first kind. The empty slot may hold three packages, not one, and the one that has passed a test so far is order structure **[H]**. The large-amplitude periodic orbits of delayed positive feedback coexist with stable equilibria but are unstable, as monotonicity requires, and orbits that pass near them are the long transients of Section 10.1 **[V-c]**.

The inverse-square flux law is the cleanest occupant of the first mechanism. It was already executed on both sides **[V]**. Check 8 computed the flux of radial fields $|F| = r^{-k}$ through spheres of radius 1 to 10 in dimensions 2, 3 and 4. The flux is independent of radius exactly when $k = d - 1$, with a spread of at most 4.441×10^{-16}, and the numerical divergence matches $(d - 1 - k)\, r^{-(k+1)}$ to 1.8×10^{-9} **[V]**. On concentric spheres the flux half of the test is close to automatic, and the Gauss content sits in the divergence table **[V-c]**. The cascade's rung 9 ran the same law in the author's language: "THE LEDGER + DIMENSION ⇒ 1/r^(d−1)". The 1/r² field's flux over 4π was 1.000000001, 1.000000001 and 1.000000000 at R = 1, 2 and 4. A 1/r field's flux grew to 3.992492 ("change appears between the spheres from nothing"). In two dimensions the 1/r field conserved flux at 1.000000000. The rung ends: "The dimension d = 3 is the remaining coordinate — a selection, flagged in the status ledger." (MN L85269–85291) **[V]**

The address of the inverse-square law is therefore (sum of local pieces | a static, isotropic, linear, source-driven field with a massless mediator | dimension d | flux through a closed surface), with value $1/r^{d-1}$ **[V-c]**. The coordinate d = 3 is not derived anywhere in the corpus **[O]**. The literature supplies selection arguments that are conditional on the law's form. Ehrenfest (1917) showed that neither classical atoms nor planetary orbits can be stable in more than three dimensions (**as reported by Tegmark**). Tangherlini (1963) showed that "the bound state postulate uniquely entails the spatial dimensionality" for the n-dimensional Schwarzschild field, from a premise that the laws "admit of a formal extension to spaces of an arbitrary number of dimensions" (**Tangherlini 1963**) **[V]**. The orbital half of the argument is elementary: a power-law force $F \propto r^p$ has stable circular orbits only if $p > -3$, so $p = -(d - 1)$ requires $d < 4$ **[V-c]**. With a massive mediator the potential is Yukawa-like and not a power law at all **[V-c]**. Any derivation of a $1/r^2$ "price of change" must therefore import both flux conservation and d = 3. The constraint chain imports the first and flags the second **[V-c]**.

10.4 What the slot predicts

The empty slot is the scheme's prediction. The runtime's "Reality Mapping" document already specifies the package it calls for: a **global_dynamics** library with methods to enumerate equilibria, continue branches, detect Hopf and saddle-node branches, detect limit cycles, compute Floquet multipliers, map basins, classify coexistence, detect hysteresis, and "compare local vs global predictions". Its test case is exactly REG's ring ("positive loop + delay + known local latch branch + observed coexisting cycle"), and its boxed rationale is "The present work has documented local admissibility. The next module must document global reachable continuation." **[S]** The address scheme adds two things to that specification **[H]**. First, the package should be built from one of two known mechanisms, index theory or gluing obstruction, rather than invented. Second, its first test is falsifiable in advance: if the arrow/coordinate distinction survives at the global layer, a resource extension will move basins without changing the set of attractor types, and a class extension will add a type. Its first two methods are now fixed by this paper's own correction. One is a persistence test: at fixed parameters, lengthen the run until the verdict stops moving, and report Occ(τ, T) rather than a basin until it

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does. The other is a sign-pattern certificate: for a monotone loop, the sign product alone decides whether a stable cycle can exist **[S]**. No attractor atlas has been built **[O]**.

11. Gödel's location: the diagonal root, with incompleteness in the reader

11.1 One theorem, two faces

REG named Gödel in its claim but emitted nothing for him. The standard single address for self-reference is Lawvere's fixed-point theorem: "In a cartesian closed category, if there is a point-surjective map φ: A ⟶ B^A, then every morphism f: B ⟶ B has a fixed point" (**Lawvere 1969**; statement as given by **nLab**) **[V]**. Yanofsky's survey shows that "many self-referential paradoxes, incompleteness theorems and fixed point theorems fall out of the same simple scheme" (**Yanofsky 2003**) **[V]**.

The theorem has two faces, and the named results divide between them **[V]**.

The **contrapositive face** says that a fixed-point-free endomap α on Y rules out any point-surjection $A \to Y^A$. This face produces the impossibility results, each with an explicit α. Negation on two values gives Cantor, Russell, Grelling, the Liar and a relativized separation $P^B \neq NP^B$. A three-valued map gives the Strong Liar. The digit map $\alpha(i) = 9 - i$ gives Richard's paradox. For the halting problem the map is partial, sending "halts" to "diverges".

The **fixed-point face** produces self-reference: the diagonal lemma, Kleene's recursion theorem, Löb's paradox, Rice's theorem, von Neumann's self-reproducing machines, and the Gödel sentence, which is a fixed point of "not provable". Tarski's undefinability can be presented from either side.

This corrects a common shortcut, which the brief for this paper also took. Gödel's incompleteness is not a bare instance of the contrapositive. It comes from the fixed-point side, and it needs more than the diagonal: a consistent, recursively axiomatized theory that interprets enough arithmetic for provability to be representable. Lawvere and Yanofsky abstract the diagonal step, not the arithmetization. The precise placement is therefore of **the diagonal core of Gödel**, not of Gödel's theorem **[V-c]**.

The address follows: (diagonal | a cartesian closed category, here Set | the endomap α on the value object Y | the representation map $e: A \to Y^A$) **[H]**. The reader slot does the work. Point-surjectivity is a reader's claim to represent every Y-valued predicate on itself, and the theorem says where that claim must fail.

11.2 The finite check

Check 6 ran the construction in Set **[V]**. It used two fixed-point-free maps, negation on {0, 1} and $y \mapsto y + 1 \bmod 3$ on {0, 1, 2}. It enumerated every map $e: A \to Y^A$ with $|A|$ from 1 to 3, which is 2 + 16 + 512 + 3 + 81 + 19,683 = 20,297 maps. In none of them did the diagonal $d(a) = \sigma\big(e(a)(a)\big)$ land in the image of e. It then drew 120,000 random maps with $|A|$ up to 12, again with zero hits. A control with σ = id, which has fixed points, landed in the image in 5000, 3748, 2876, … down to 26 of 5000 trials, so the fixed-point-free hypothesis is what does the work, not the counting. The case Y = {0} is the one place where point-surjective maps exist, and there the only endomap, the identity, has a fixed point. That is exactly what the theorem requires: the case confirms it rather than contradicting it.

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Two limits apply. The column counting point-surjective maps is empty by counting alone, since $|Y|^{|A|} > |A|$, so the evidence is the diagonal-escape column. And no finite test reaches the theorem's categorical statement **[V-c]**.

11.3 C1 is Lawvere's hypothesis, and the corpus puts incompleteness in the reader

The author's first constraint, C1 ("all things must change"), read as "the change map has no fixed point", is exactly the hypothesis of the contrapositive face. It does not need aperiodicity, since negation is a 2-cycle **[V-c]**. The Co audit reports that of the four maps on a two-state set, "the swap is the only C1-legal map" (review of the Co Audit Trail; the underlying document is not in the uploads) **[V-c]**. At the binary base, then, C1 is Lawvere's canonical α. A world that obeys C1 there cannot contain a reader that represents all of its own binary predicates **[V-c]**. Two conditions attach to that formal consequence. The change map must be a morphism of the ambient category (computable, or continuous, as the setting requires), and states must be global points **[V-c]**.

There is also a trap. In pointed domains, where a divergent value ⊥ exists, every endomorphism has a fixed point, and Yanofsky's halting instance works by treating divergence as a value. "No terminal state" and "no fixed point" are therefore different hypotheses, and treating them as one reverses the logic **[V-c]**.

The assistant's chat derivation of the incompleteness limit from C1 checked, among other things, every fixed-point-free map on 2 to 6 points: 16,739 of 16,739 generate a nontrivial cycle, an exhaustive count equal to $\sum_{n=2}^{6}(n-1)^n$ **[V-c: chat-reported, recount confirmed]**. The same assistant labelled the mathematics "classical — cite, don't claim" while calling the derivation from one ontological axiom the author's own (MHC msg100) **[V-c]**.

The notebooks take a consistent position on where incompleteness lives: in the reader. "What a Cut Must Do to Make a Thing" says the declared set "**is** the quotient $\mathbb{Z}/2^n \to \mathbb{Z}/2^{n-1}$. The blind bit is not a defect in the predicate. It is the cut the predicate *is*. Placing the incompleteness in the object would be the mistake here; it is in the reader, and the reader is the predicate." **[V as text]**. The carry-bit paper has "Place the incompleteness in the reader; the odd factor is where it sits." **[V as text]**. "What a Cut" also proves that "The update is a function forward and **not a function backward**. A reader can close a cut from its state alone; it cannot open one." (all n) **[V]**.

The word "Gödel" appears nowhere in the merged notebook. The Lawvere reading of the adder and the cut lives in chat (MHC msgs 79–100), and a reviewer phrased the consequence as "Incompleteness is not abolished; it is the recurrent production of a new cut whenever a reader becomes expressive enough to read itself" **[S]**. The cascade's rung 10 reports "correct self-predictions: 0 / 10,000. the reader NEVER closes", but its code sets truth = 1 – prediction. It is a diagonal construction executed, and it cannot come out otherwise; it is not a test **[V as construction; not evidence]**.

11.4 FIX is the open question at this address

The Co Audit Trail, known only through a review, reduces its primitives to four operations, DISTINGUISH, ACCUMULATE, COMMIT and FOLD. It reports that FOLD ≠ FIX: FOLD oscillates $0 \to 1 \to 0$, whereas a contraction reaches a fixed point **[V-c]**. It leaves open "Ω_3: Is self-reference a fifth irreducible primitive?" **[O]**. The proposal for FIX is to "make the result of a transformation available as a condition of that transformation's own later continuation", moving from $F(F(x)) = x$ to $P \simeq T(P)$, and its boxed contrast is "FOLD gives return; FIX gives self-maintained identity." **[S]**

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Kleene's recursion theorem supplies such behavioural fixed points without state fixed points. A quine run in chat reproduced its own bytes exactly, and the gloss was "The quine's fixed point is in behaviour, not in state … C1 holds for the bytes, and the recursion theorem holds for what the bytes do." **[V-c]**. On the address reading, C1 lives on the contrapositive face of the diagonal root and FIX on its fixed-point face **[H]**. Whether the diagonal is a fifth root, or a reading of an existing one, is the registry-level form of Ω_3, and it is open **[O]**.

12. The constraint chain closes, and location is where physics starts

12.1 Ten rungs, each C1 on one more degree of freedom

The author's chain, recorded as a specification on 13 July 2026, runs from "C1: all things must change" through "C2: all things must have infinite degrees of freedom for the potential to change", "Law 2: no matter may hold a location without the potential to be moved from it", transformation, the price as a gradient, "mass x acceleration = cost to transform", direction and collision, to "C3: … all change is equal" and "Law 3: there is no end". The executed cascade states its method in one line: "Every section below is C1 evaluated on ONE degree of freedom. Run first, write after." (MN L84568–84586) **[V as text]**. Table 7 gives each rung's executed evidence and the classical result it re-runs. The notebook credits most of those results by name.

Table 7. The constraint chain rung by rung: claim, executed evidence and classical content.

Rung	Claim	Executed evidence	Classical content
1 C2	C2 is C1 applied to the potentials	all 15,625 = 5^6 fixed-point-free maps on 6 states freeze their image chain within 4 steps; successor on $\mathbb{N}$ never freezes (image floor 10^6 after 10^6 steps)	descending chains on finite sets
2 location	no location may be held; the hold has a diverging price	E_0 = 0.499999319221; $\sigma_x\sigma_p$ = 0.4999993192; hold tax 1.25×10^{11} at $\sigma_x = 10^{-6}$	uncertainty, zero-point energy
3 transformation	the new location inherits the law; inheritance is conservation	momentum drift 2.37×10^{-14} over 200,000 ticks; with a pinned well, 1.20	Noether (credited)
4 gradient	a single value cannot prove change	V → V + 1000 moves trajectories by 4.36×10^{-9}; closed-loop work: gradient field 6.10e-19, curl field 2π	exactness of conservative forces
5 inertia	m·a = cost to transform	$\max\lvert ma - F\rvert = 2.51 \times 10^{-7}$ for m = 1, 2, 5; (m, G) → (10m, G/10) bit-identical	equivalence-principle scaling (credited)
6 wash	balance reads as absence	net force 1.13×10^{-15} in a 9 × 9 × 9 lattice; remove one mass and it reads (−1, 0, 0)	shell theorem (credited)
7 direction	direction is forced and ledgered	angular-momentum drift 1.93×10^{-13}	Noether, rotational face
8 collision, C3	all change is equal	Δp = ∓1.50000005, ledger -8.88×10^{-16}; energy error 9.99×10^{-9}	elastic collision
9 field geometry	ledger + dimension ⇒ $1/r^{d-1}$	flux/4π = 1.000000001 for 1/r² in 3-D; d = 3 "a selection"	Gauss (credited)
10 Law 3	there is no end	minimum speed 0.999987 over 10^6 ticks; 0/10,000 self-predictions (by construction)	Poincaré recurrence (credited)

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The chain closes on itself. The rung-10 argument is: "'no end' + FINITE states would force eternal recurrence (Poincare, credited) — a frozen orbit, the rung-1 violation. So Law 3 re-forces C2 … The chain has no loose end BECAUSE it has no end." (MN L85294–85318) **[V as argument; the recurrence step is credited, not executed]**. Figure 8 draws the loop. It closes at C2, not at Co, so Co and C1 read as the entry into a cycle.

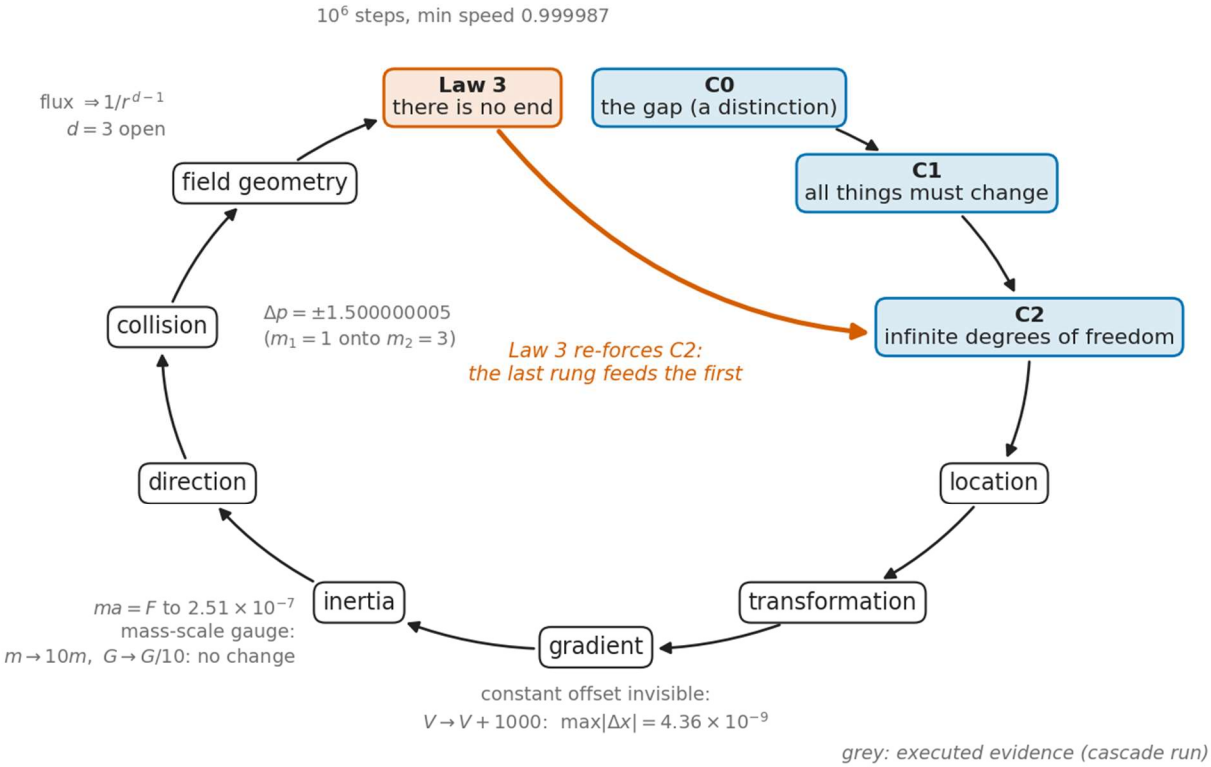


Figure 8. *The closed constraint chain. From Co (a gap across which change can be measured) and C1 (all things must change), each rung applies the change constraint to one more degree of freedom: C2 (an inexhaustible state space), location (no location may be held), transformation, gradient, inertia, direction, collision and field geometry, ending at Law 3 (there is no end). Law 3 re-forces C2 (thick arrow), so the chain closes on itself at C2 rather than at Co. The grey notes are executed evidence from one cascade run; the one coordinate that remains open is the dimension d = 3. The return arrow is an argued implication crediting Poincaré recurrence, not an executed step.*

Three cautions travel with the chain.

- What is original is the ordering and the reading, one degree of freedom per rung, not new physics. The external review said as much. It rated the cascade highly but added that it "does **not** by itself establish that the physical universe must obey the entire NEXUS construction, nor does it derive the observed value of spatial dimension d=3" **[V-c]**.
- The corpus holds three different operational "Law 3"s. The first is the cascade's "no end". The second is the arena's candidate, that loss happens only where two occupancies are forced into one location, which the arena itself calls "a reading of the measurements, not a theorem". The third is the Executable Record's recurrence test, in which a projection recurs while the full state does not: the closest loss match, steps 960 and 1176, still differs in weight space by 3.145, about 55% of the typical distance. The three are compatible in spirit, but they are not one statement and should not be merged **[V-c]**.

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- The logical face of rung 10 proves nothing, because it holds by construction (Section 11.3) **[V]**.

12.2 Why location is the rung where physics starts

Read the chain against the registry and one rung stands out **[H]**. Rung 1 is a theorem about maps on finite sets. It has no units and needs no space. Rung 2 is the first rung whose executed output is a physical constant: a ground-state energy of one half, $\sigma_x\sigma_p$ = one half, in units where ħ = 1. It is also the first rung that needs a pair of observables that do not commute, which is to say the first that calls the Gram root. Everything after location is location's inheritance worked out. A law that is the same at every location conserves momentum; pin the well to one place and conservation fails by 1.20 **[V]**. The price is a gradient because a constant offset cannot be read. Inertia prices the change of location. Collision is two movers demanding one place. The flux law is the ledger spread over the surface around a location.

On this reading, location is where the change constraint first meets a continuous coordinate with a price attached, and that price is Planck's: Law 2 is the uncertainty relation (Section 7.5). The arena reached the combinatorial version independently: the one operation that destroys structure is "the prohibition on two occupancies sharing one location", and "Its content is the **location** of the loss, not its size." (MN L109040–109067) **[V]**. "Location" thus has two executed senses in the corpus. One is physical: the hold tax, and inheritance as conservation. The other is combinatorial: exclusion. "Address" is the reader-side notion, as in loc(S), self-addressing and access geometry. The bridge between them, which the specification lists as "a reality-level derivation of spacetime geometry from access geometry", is open **[O]**.

13. Mathematics has location

If named laws are boundaries at addresses, the constants that appear in them are coordinates of those boundaries. That is the most this paper can mean by "even math has location" without passing into metaphysics. The executed content of each case below is that a number is the value of a root test at a specific coordinate **[V]**. The reading of the number as a location is marked **[H]**.

Integers are addresses on the successor orbit. Rung 1 of the cascade shows the contrast. The cyclic successor mod 12 moves every state while its potentials never move, whereas the successor on $\mathbb{N}$ never freezes **[V]**. An integer names a state on that orbit. Relative to a period B, $n = qB + r$ splits the address into a winding count q and a phase r, and a carry is a phase slip. The chat table that first set out the correspondence pairs digit count with winding, remainder with "phase mismatch, strain, or detuning", and carry with "phase slip" **[V-c chat; H reading]**.

The carry itself has an exact mathematical location. It is a group 2-cocycle, $c(a, b) = 1$ if and only if $a + b \geq 10$, which classifies the non-split central extension $\mathbb{Z}/10 \to \mathbb{Z}/100 \to \mathbb{Z}/10$, and iterating it gives the 10-adic integers (**Isaksen 2002**; **nLab: carrying**) **[V]**. Because the extension does not split, no bookkeeping removes carrying **[V-c]**.

The corpus adds an executed location result in arithmetic. A cut S of an N-bit word survives regrouping into base-2^k digits exactly when k divides $\mathrm{loc}(S) = \gcd(\partial S \cup \{N\})$: 25,904 pairs, zero mismatches. At N = 16, 65,280 of the 65,536 cuts are nameable only at the finest rung, and 2 are universal (Carry Capacity §8) **[V]**. On the reader side, "Two Quotients of One Machine" proves that the minimal Myhill–Nerode state count after reading a set of positions "depends only on S — never on the order used to reach it", which it summarizes as "Memory is a cut functional, not a history functional" **[V]**. The carry-bit paper adds: "The carry *bit* is a property of the bottom-anchored cut, not of addition." **[V]** A least-significant-first adder is a

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two-state transducer whose state is the carry, and by Myhill–Nerode that bit is the minimal memory. Isaksen's cocycle and Nerode's minimal reader are therefore one object read two ways (**Myhill–Nerode theorem**) **[V-c]**.

π is the half-turn of the phase circle. In the resolver, π is not a number that happens to appear. It is the coordinate value of Φ at which a negative loop closes: the latch lives at the zero turn, the clock at the half-turn, and the full turn is preempted **[V]**. Reading π as "the place where counting lives" is interpretation **[H]**.

e is the diffusive exchange rate, one neper per radian. On a diffusion line, magnitude and phase are the same number: $|H| = e^{-\theta}$ **[V]**. The constant e is the base in which the two currencies trade one for one, so e^{π} is the price of a half-turn paid in diffusive currency **[H]**.

The secant is a per-stage share. N identical stages each carry π/N of the phase, and $\sec(\pi/N)$ is each stage's share of the magnitude bill **[V]**. As N grows every share tends to 1, which is the pure-delay limit. Griffith's 8 is the whole bill for three shares **[V]**. "Share" is a reading **[H]**.

5/6 is a point on a frontier. Bužek–Hillery's 5/6 is F(1, 2) on the Gisin–Massar surface. It is also the symmetric midpoint of the asymmetric frontier, whose computed point at equal weights is (0.833332, 0.833335) and whose ends are (1, 1/2) **[V]**. A constant, on this reading, is where a frontier crosses a symmetry line **[H]**.

The BBP 8k + r reading is not in the corpus. The corpus uses the Bailey–Borwein–Plouffe formula only as an example of "access without traversal", addressing a digit without walking through its predecessors. A reviewer capped even that: BBP should "remain an example—not the foundation of the general claim" **[S]**. The reading of $n = 8k + r$ as winding k and phase r was searched for in the merged notebook, all three chat exports, the attachments and both earlier documents, and it is not there **[O]**. The literature explains the 8. The series $\pi = \sum_{k\geq 0} 16^{-k}\,[4/8k + 1 - 2/8k + 4 - 1/8k + 5 - 1/8k + 6]$ comes from expanding $1/(1 - x^8)$ in an integral taken to $1/\sqrt{2}$, so the period 8 is set by $16 = \left(1/\sqrt{2}\right)^{-8}$. It is a property of the base and the integral, not a special property of π's digits (**Bailey, Borwein & Plouffe 1997**; coefficient check in the literature notes) **[V-c]**. Digit extraction runs in nearly linear time with virtually no memory. It shows that a digit can be addressed by position, not that it can be read at no cost, and it says nothing about whether π is normal **[V-c]**. A winding-and-phase reading of 8k + r would be **[H]** at best; reading the 8 as a fact about π would be wrong **[R]**.

14. Prior frameworks already locate laws; this scheme must predict

Giving laws a location is an old ambition, and most of the address scheme's ingredients have precedents that must be credited rather than rediscovered.

Klein's Erlangen program (1872) located whole geometries by their transformation groups. It stated, in so many words, the clause that a larger group hides structure: "If the principal group be replaced by a more comprehensive group, a part only of the geometric properties remain unchanged." It also stated that two differently named theories can be one location, since "The theory of binary forms and the projective geometry of systems of points on a conic are one and the same" (**Klein 1872, Haskell translation**; **Math. Ann. reprint**) **[V]**. The group acts like a reader, because it decides which distinctions count. Neither "a reader change hides" nor "two named results share an address" is new.

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Tonti's classification diagram is the strongest precedent for giving individual laws a coordinate. Every physical variable is placed by its category (configuration, source or energy), its space element (point, line, surface or volume), its time element (instant or interval) and its orientation (inner or outer, the primal or dual cell complex). Every equation is classified by its link: vertical links are topological, and horizontal links "are material or constitutive equations" (**Tonti diagram, Encyclopedia MDPI**; **Tonti, "Sulla struttura formale delle teorie fisiche"**) **[V]**. Discrete exterior calculus builds the same split into simplicial complexes (**Hirani 2003**; **Desbrun et al. 2005**) **[V]**. For classical field theories, "a named law is an address" is prior art. The frequently cited date of 1972 for Tonti's English monograph could not be confirmed **[O]**.

Noether (1918) gave every conservation law the address of a symmetry, and her second theorem made gauge redundancy the precedent for transformations that stay in the formalism while only invariants are observable (**SEP, Symmetry and Symmetry Breaking**) **[V]**. The cascade's rung 3 re-runs exactly this and credits it. **Constructor theory** recasts laws as statements about "possible and impossible physical transformations", and it treats no-cloning as a property of superinformation media (**Deutsch 2013**; **Deutsch & Marletto 2015**) **[V]**. It supplies transformations as primitives and a notion of distinguishability close to a reader, but it gives laws no coordinates. The arrow/coordinate split itself has a close precedent in quantum resource theories, which divide operations into free operations and resources (Chitambar & Gour 2019) **[V-c: citation not re-checked]**; the criterion here differs in testing whether the attainable set strictly grows. **Lawvere** made laws of motion objects in a topos (**nLab: topos of laws of motion**) **[V]**. **Bourbaki's** three mother structures (algebraic, order, topological) are the canonical precedent for "a few roots". Bell's assessment is also a warning: "the concept of mathematical structure as such plays a very minor role in Bourbaki's development" (**Bell**) **[V]**. An address scheme that only relabels does no work.

Four further comparisons are cautions rather than support. Wigner found the effectiveness of mathematics "something bordering on the mysterious" and allowed that "no union of the two theories can be found" (**Wigner 1960**) **[V]**. Tegmark's Level IV makes physical reality one mathematical structure among all of them (**Tegmark 2008**) **[V]**. Wolfram's ruliad has observers select "some 'slice of the ruliad'" and holds that "physics and mathematics are at their core the same thing" (**Wolfram 2021**) **[V]**; it is the closest precedent for a reader, and it is an essay. Wheeler's "it from bit" makes the observer participatory (**Wheeler**) **[V]**. "Laws as software" is itself prior art: Kantor (1982) and Davies (2007), who wrote of "the laws of physics in terms of software, as a grand cosmic computer program" (**Davies 2007**; **SEP, Computation in Physical Systems**) **[V]**. The same SEP entry supplies the objection any such scheme must answer: "It doesn't follow that the modeled systems are computing devices", and "There is little positive evidence to support ontic pancomputationalism." **[V]** Literal named-law addresses exist today only in formal libraries such as Physlib, formerly HepLean (**Tooby-Smith 2025**) **[V-c]**. These organize human knowledge and make no claim about nature. This paper follows the runtime specification's own caution and treats "library" as an architecture for organizing laws, not as a claim that the universe executes code **[H]**.

What survives as new is narrow and checkable **[H as a novelty judgement]**. Each named law gets an explicit reader coordinate. A resolver computes addresses and refuses when a bill cannot be paid. The roots are counted, and the count can be falsified by collapse. Two placements, Planck at the Gram root and Laplace at the edge of the cloning surface, are made here. Above all, the empty addresses are used as predictions, the way the periodic table used its gaps. The scheme makes six predictions.

1. **The next package is global.** It will be built from index theory, a gluing obstruction or order structure. For sign-structured loops the order-structure package already exists in the literature and

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has passed one test here (Section 10.1). At the global layer a resource extension should move basins while a class extension adds an attractor type (Section 10.4) [O].

2. **Unvisited root-1 addresses have computable values.** A loop closed through an open-ended finite RC line should need a gain of $\cosh\pi \approx 11.59$. A detuned resonator should need $2\zeta(\rho + 1/\rho) + 4\zeta^2$. Unequal stages should need more than $\sec(\pi/N)$. A telegrapher line with shunt loss is not bounded by e^{π}. The literature check derived and numerically checked the first two [V-c]. None has been run through `resolve()`, which needs a new crossing kind for each [O].
3. **Roots 3 and 4 may be readings of root 2.** If they are, the registry has two roots, not four (Section 8) [O].
4. **Erase is a price.** $kT\ln 2$ should appear as a reset cost that vanishes with temperature, for instance as the hysteresis area of the latch [O].
5. **Gate and Merge must have addresses.** If no existing root emits them, the registry needs a new root [H].
6. **d = 3 cannot come from the flux package.** Inside it, d is a coordinate. Any derivation of three dimensions must come from a different root, such as the stability selection of Ehrenfest and Tangherlini [H].

15. Method rules and the complete error log

The error log is reported here as a result, not an appendix. Every notebook after VRT refuted or withdrew something, and three of the programme's lasting objects came out of errors. The closed form for $a_c(n)$ came from a failed search. The form $\exp\left(\pi\sqrt{1 + \mathrm{Da}}\right)$ came from a phase-wrapping bug. The gain-first design formula, used in every later ring simulation, came from an overflow [V]. E20 was found in review of this paper. About 7 of the first 19 items appear in the notebooks' own corrections sections. The rest were reconstructed from the session transcript and run logs, so Table 8 is a reconstruction [V-c].

***Table 8.** The complete error log, reconstructed from the notebooks, the session transcript and the run logs.*

ID	Error	Symptom → cause	Fix → rule	Status
E1	search failure (VRT)	root-finder fails at small a → the critical value was below the bracket	derive $a_c(n)$ from $\lvert f' \rvert = 1$ → derive, don't search	[V]
E2	bracket failure, overflow (OB)	target gain below 1; x^n overflows	closed-form gain-first design, n = 1.18G → set the target by construction	[V]
E3	timeouts, kernel deaths (VRT, OB, LOAN)	2-minute limit; buffered output lost when runs were killed	vectorize, trim, split runs, log to file → time each cell first	[V]
E4	stale count (OB)	table says 40/40; cell prints 30 of 30	none in the notebook; corrected here	[V]
E5	off-by-one stage index (OB)	two rings the spectrum calls stable oscillate in integration → stage i applied to x_i instead of stage i+1	residual column → "The disagreement was the only reason the bug was found."	[V]
E6	wrong branch (OB)	5 of 8 agree → negative-loop Φ = π applied to a positive loop	choose Φ by loop sign → the Φ table	[V]
E7	own prediction refuted (OB)	delayed 2-ring predicted to clock; it latched	preemption: the latch opens first	[F]
E8	slow transient read as a cycle (OB, PRICE)	swing 1.44e-1 at T = 300 → decays to 2.4e-11 at T = 4000	run length $\propto 1/\lvert\mathrm{Re}\,\lambda_{\max}\rvert$	[V-c]

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ID	Error	Symptom → cause	Fix → rule	Status
E9	biased sampler (OB)	M_c = 64, the bracket edge → draws favoured x > y	symmetric Dirichlet sampling → "A test that finds nothing is not evidence of nothing"	[V]
E10	patches to wrong cells (VRT, OB, LOAN)	stale notebook still "executes cleanly"	rebuild from the build script; print the index map first	[V]
E11	coarse spectral resolution (LOAN)	Welch peak 0.2441 at every Ω	time-domain decorrelation of the order parameter	[V]
E12	principal-branch phase (CEIL)	**np.angle** wraps 4 rad to 4 − 2π; $G_{\mathrm{crit}} = \infty$	analytic branch → the closed form $\exp\left(\pi\sqrt{1+\mathrm{Da}}\right)$	[V]
E13	**copy.py** shadows the standard library (CEIL)	circular-import error in scipy	rename the file	[V]
E14	one-sided fidelity (CEIL)	identity map scores 1.0 at every d_a	score both marginals; identity baseline 0.75 → "results that were too good"	[V]
E15	Monte Carlo drift at N = 50 (SIG)	0.985819 against 0.980769 → uniform sampler misses a concentrated peak	diagnosis stated, never re-run	[V-c]
E16	own prediction refuted (LOAN → PRICE)	one second-order stage predicted to oscillate; order 2 never does	count total dynamical order	[F]
E17	scoped claims (VRT, OB)	"counting costs one more axis"; parity sufficient	scoped to the delay-free class; parity re-derived from the loop sign	[V]
E18	NaN, lost output (OB)	0/0 at a degenerate fixed point; buffered stdout under timeout	skip degenerate states; **python3 -u** and log files	[V]
E19	language corrections from reviews	Φ "conserved"; resources "interchangeable"; "no larger class"; Choi saturation	accepted in LOAN, CEIL, SIG and in session	[V]; Choi items [V-c]
E20	finite-horizon occupancy read as a basin (REG)	fraction oscillating at T = 500 rises 0 → 0.75 with τ → the ring is a monotone positive loop; the oscillations are long transients	horizon test: at fixed τ the count falls as the run grows (τ = 5: 37 → 20 of 48; τ = 2: 9 → 0) → report Occ(τ, T) until the verdict is horizon-stable; check the sign pattern first	[R]

The rules that came out of the log are the programme's method, and each first appears at a traceable place [V].

M1, audit the reader. It was boxed in SIG c12: "When output is too good, too absolute, or violates a known limit: audit the reader before revising the world." ARROW gave the procedure: "ask which quotient, branch, marginal or region the measurement discarded — before revising the model." The five-term form ("quotient, branch, marginal, measure, or concentrated region") appears only in a review, not in any notebook [V-c].

M2, integration time scales with proximity to a bifurcation. Near a threshold the decay rate vanishes, so run length must scale with $1/|\mathrm{Re}\,\lambda_{\max}|$ "or a slow transient will pass for a cycle." E20 extends it: near a long-lived transient the relevant time is the transient's lifetime, so a verdict must be checked for stability under a longer run.

M3, verify the object. Hence the residual column in OB.

M4, a null result measures the test. Sampling must respect the symmetry of the question.

M5, fix predictions and closed forms before the run, and use them only to check. "Nothing here is told where the threshold is."

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M6, derive instead of searching.

The corpus adds rules of the same kind. The cascade's "Run first, write after". The Executable Record's "Kills are first-class". The consolidating arena paper's "prove downward, interpret upward, never prove upward by interpretation". Carry Capacity's "A law found in a corner cannot be distinguished from a law about the whole family by inspection, by review, or by agreement. It can only be distinguished by moving a parameter." And the Co audit's self-count: "9 retractions; 3 failed demonstrations; 8 of 12 corrections traced to prose being written before the corresponding computation returned", with the rule "Measure the relation, not the value, and at the right level." [V-c]

The errors sort into four classes. Reader failures (five: index, measure, branch, marginal, concentrated region) were caught by impossible numbers. Run-length failures were caught by disagreement near a threshold, and once, in E20, by theory from outside the programme. Theory errors (the wrong branch and two refuted predictions) were caught by integration, and each became a result: the Φ table, preemption and the order bound. Infrastructure errors were caught by exceptions rather than numbers. ARROW's "None found by reading code; all found by a number that could not be right" is true of the first class only [V-c].

In address terms, each reader failure was an unintended move in the reader slot [H]. The phase branch is a principal-branch reader. The fidelity marginal is a one-output reader. The sampler is a biased measure. The concentrated region is a uniform reader of a peaked integrand. The address scheme predicts where its own errors will occur, and the log agrees.

16. What is not new, and every correction

Every named result in this paper is classical, and the programme's notebooks say so themselves: "What this notebook adds is the resolver, the root count, and the location of the hole" (REG c8) [V]. Table 9 lists the primary sources and how far each was verified in this research. [V] means the content was read; [V-c] means only the bibliographic record was confirmed.

Table 9. What is not new: primary sources and how far each was verified.

Named result	Primary source	Verification
Nyquist criterion	Nyquist 1932, BSTJ 11, 126	[V-c]
Barkhausen criterion	Barkhausen, *Lehrbuch der Elektronen-Röhren* vol. 3 (1935), via secondary	[V-c]; no page located
Secant condition	Tyson & Othmer 1978; Thron 1991; Sontag 2006	[V] Sontag; others [V-c]
Griffith's n > 8	Griffith 1968; Goodwin 1965	[V-c]; text not read
Toggle and repressilator	Gardner, Cantor & Collins 2000; Elowitz & Leibler 2000	[V-c]; Elowitz–Leibler unverified
Toggle threshold $a_c(n)$	attributed to Cherry–Adler by the registry	[O] attribution unchecked
Positive/negative circuit rules	Thomas 1981; Gouzé 1998; Soulé 2003	[V-c]
Delay stability	Hayes 1950	[V-c]
Hopf bifurcation	Hopf 1942, translation	[V-c]
No-cloning	Wootters & Zurek 1982; Dieks 1982; Park 1970	[V-c]
Probabilistic cloning	Duan & Guo 1998	[V] full text

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Named result	Primary source	Verification
Universal cloning, 5/6	Bužek & Hillery 1996; Gisin & Massar 1997; Werner 1998	[V-c]
Estimation (N+1)/(N+2)	Massar & Popescu 1995	[V-c]
Dilation, Choi rank	Stinespring 1955; Choi 1975	[V-c]; theorem re-derived
Landauer bound	Landauer 1961; Bérut et al. 2012	[V-c]
System-size noise	van Kampen 1961; McKane & Newman 2005	[V-c]
Uncertainty relations	Kennard 1927; Robertson 1929; Schrödinger 1930	[V-c]; placement derived
Diagonal theorem	Lawvere 1969; Yanofsky 2003	[V]
Carry cocycle	Isaksen 2002	[V-c] via nLab
Digit extraction	Bailey, Borwein & Plouffe 1997	[V] abstract
Dimension selection	Tangherlini 1963; Ehrenfest 1917	[V] Tangherlini; Ehrenfest via Tegmark

What might be new is short, and even the candidates carry caveats [H as a judgement].

- **The resolver and the root count.** A single interface emits nine published loop thresholds from class descriptors, and the registry has four roots with one empty slot.
- **The diffusive transport axis.** The e^{π} threshold, its decay form and its telegrapher sweep were not found in the literature. That search was at title level.
- **The Laplace link.** No published source was found.
- **Two placements.** Planck's floor at the Gram root, beside no-cloning; and the Griffith/ring-oscillator identity as one address.
- **In the carry family.** The m-operand, base-b splitting lemma, whose novelty the corpus's own assistant flagged as pending a check of Wegener's monograph on branching programs [O].

None of these absences of prior art is evidence of novelty.

Table 10 consolidates every correction that placing the results at exact addresses forced, or that the record shows.

Table 10. Every correction that placing the results at exact addresses forced.

Claim as it stood	Correction	Marker
$a_c(2) = 4$ (this paper's brief)	$a_c(2) = 2$; 4 is the supremum of the n = 2 loop gain	[F]
sec(π/N) for any N-stage ring	tight only for identical stages; higher otherwise	[V]
Griffith: n > 8	an infimum approached as a $\to \infty$; $n_{\min}(a) - 8 \approx 8a^{-8/9}$; "a ≤ 1 never oscillates" false for $n \gtrsim 2.39 \times 10^4$	[V-c]
$\exp(\pi\sqrt{1+\mathrm{Da}})$	only with $\mathrm{Da} = kT/\pi^2$; with Da = kT the threshold is $\exp(\sqrt{\pi^2 + kT})$	[V]
order ≥ 3 is a gate	class-relative: delay-free lags and resonators, no self-activation	[V]
bare Duan–Guo matrix for any state set	exact for two states only; sufficient for three or more; rate jumps at x = 1	[V-c]
5/6 and (N+1)/(N+2) emitted by the Gram root	reached by optimization and Monte Carlo, not from the condition	[O]

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Claim as it stood	Correction	Marker
saturation point = Choi rank	d_a 1 → 2 is a class extension; completion checked at six points; ARROW c5 and the claims ledger should read	[V-c]
global onset "a factor of about eight"; "basin fraction"	the oscillations are long transients: at fixed τ their count falls as the run grows (τ = 5: 37 → 20 of 48 from T = 500 to 4000; τ = 2: 9 → 0); the measured quantity is finite-horizon occupancy, not a basin; its first appearance at T = 500 is 6 to 12 times the local value	[R]
no-cloning and Duan–Guo, "two literatures, sixteen years apart"	a unifying presentation, not a discovery: Duan and Guo generalized no-cloning, and the p = 1 case reduces to it by construction	[R] as framing
"dropping one input climbs (N+1)/(N+2)"	(N+1)/(N+2) is the state-estimation fidelity (the M → ∞ edge); an N → N+1 cloner reaches ((N+1)² + N)/((N+1)(N+2))	[V]
carry cut law transmit(g) = ⌈m/b^g⌉, impose(g) = m − ⌈m/b^g⌉ + 1	chat-only (MHC msg5; the RHA synthesis); the sum m + 1 holds by definition; executed counterparts are the forgetting-gap lemma (220 triples, 0 mismatches) and the ⌈log_b m⌉ threshold (11/11); the product law fails at m = 3, b = 2 (42 of 128 cuts at width 7)	[V-c]
H = π/9 "the unique fixed point" (RHA/Helix synthesis; April 2026 cut-density paper)	$W_0(1/2)$ = 0.351734 lies 0.76% from π/9 = 0.349066, a near-coincidence; by the paper's own definition H = 2.2100; the factor 24 names a group of order 12; π²/1944 = 0.005077, not 0.005074; the Bianchi step divides by a quantity assumed zero; nothing in the paper was executed, and its own status boxes say the SHA-256 comparison and the H_0/S_8 fits were not performed	[R] not promoted
θ = ½(π/e) ≈ 0.35 and an RH "proof by collapse" (December 2025 Gödel document)	½(π/e) = 0.578; the proof is not one; moving to a meta-layer produces a new undecidable sentence, exactly as Lawvere requires	[R] not promoted
"hidden ≠ destroyed" as a notebook result	specification wording; the executed statements are "Symmetry filtering ≠ destruction" and "projection erases visibility, not structure"	[V]
$\mathcal{H}^1$_G = 0 (consolidating arena paper)	superseded: the bull graph has $\varepsilon_{\text{harm}} = 1$	[R]
"the reader NEVER closes", 0/10,000	holds by construction (truth = 1 − prediction)	[V] as construction only
the verb set is substrate-independent	rests on at most 7 of 9 verbs; Gate and Merge untested; Erase uncomputed	[V-c]
Gisin–Massar verified	attainment verified; optimality from the literature	[V-c]
OB's "T = 900 … 9×10⁻³"; OB's missing item 3; VRT's "clock side 17 of 18"; PRICE's "third notebook in a row"	two runs conflated; an item deleted by a patch; a mislabel; only two instances found	[R] as wording

The two earlier .docx papers are kept out of the registry entirely. "Gödel Curvature and Harmonic Recursion" (December 2025) has no author line, no ORCID and no licence, attributes its Gödel reading to "Grok", and executes nothing. "Quantum Cut-Density as the Origin of Spacetime Curvature" (April 2026) carries the author's ORCID but no licence line; its own status boxes admit that the SHA-256 comparison "has not yet been executed" and that the H_0/S_8 fits "have not yet been performed". The checks recorded above show that their headline constants do not hold, and no constant from either document is promoted [R].

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17. Open specifications

Six items define the next stage, in the order the evidence makes them urgent.

1. **The Global Attractor Library.** The specification exists (Section 10.4), and its first two methods are now fixed: a horizon-persistence test, run once (Section 10.1), and a sign-pattern certificate for monotone loops. No attractor atlas has been built, and the arrow/coordinate criterion has not been tested at the global layer **[O]**.
2. **Root collapse.** Test whether dilation and counting measure are the rank and trace readers of Gram positivity. The registry then has two roots or four **[O]**.
3. **FIX and Ω_3.** Is self-reference a fifth irreducible operation, and is Lawvere's diagonal a root or a reading? The audit's Ω_5, whether a centre cell needs FIX while orbiters need only the four operations, rides with it **[O]**.
4. **d = 3.** It is a coordinate inside the flux package. No derivation exists anywhere in the corpus **[O]**.
5. **Which extensions preserve a class.** This needs a general criterion for when an extension is a resource coordinate and when it is a class extension (the specification's `classify_extension`), and a proof of the set-completion conjecture $d_{\min}(\mathcal{F}) = \max_{x\in\mathcal{F}} \operatorname{rank} J_x$ **[O]**.
6. **The Erase price check.** Compute Landauer's cost where the programme said it lives, as the hysteresis area of a switching latch, and confirm that it vanishes with temperature. That would place $kT\ln 2$ by computation rather than assertion **[O]**.

Behind these sit smaller items **[O]**:

- resolvers for M_c, for Gate and for Merge;
- a crossing kind for finite RC lines, to test the cosh π prediction;
- literature checks for the Laplace link, the Cherry–Adler attribution and the Wegener question;
- the uncertainty placement for mixed states and unbounded domains;
- non-Hill transfer shapes and asymmetric rings under noise;
- the reader bridge's open question, "Is there any reader-side object that is not a pullback?";
- the specification's bridge from access geometry to spacetime geometry;
- a single dispatcher that resolves an address at any root.

18. Conclusion

The answer to "where do named laws actually live?" is: at addresses in one registry, and the registry is smaller than the list of names and larger than one idea. A named law is the boundary value that a root test returns at a coordinate inside a class, as a particular reader reads it. Its name records who compiled that class first, which is why the same address keeps turning up in different literatures under different names: Griffith's 8 and the ring oscillator, no-cloning and Duan–Guo, Laplace and Massar–Popescu, Law 2 and Robertson–Schrödinger. The named results are not random, and the reason is checkable: a few conditions of genuinely different mathematical kind emit the named values, and one function reproduces them to better than one part in 10^{10} and refuses when a phase bill cannot be paid **[V]**. The author's intuition that "they all have location — even math" survives in a precise form. π, e, the secant and 5/6 are coordinates of boundaries, and that reading organizes the evidence without adding to it **[H]**.

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What changed in the course of the work is the claim of completeness. The evidence supports "one system", but not "one root", and not yet "complete". The registry's most valuable entry is the one it lacks. Both corpus families show the same gap between what a local condition certifies and what the global system does: a Hopf point that is right about one state and silent about where the orbits go, and a reader that "never had access" to the loop structure it seems to lose. On the loop side the gap turned out not to be empty of theory. The apparent cycle was a long transient, exposed by lengthening the run, and the package that predicts it, order structure, was already in the literature. The scheme's sharpest prediction survives in a stronger form. The next library is a global one, built from index theory, a gluing obstruction or order structure, and the first global question is not "how large is the basin?" but "does the verdict survive a longer run?" Location is also where the constraint chain turns into physics. If that is right, the global library is where the chain's "no end" will either close into a system or show that the system has further roots [H].

Acknowledgment

Computation, drafting and literature checks for this paper were done with Claude (Anthropic). This includes the nine placement checks, the figure scripts and the dense re-run of the delayed ring. The loop/channel programme was shaped by external reviews uploaded between notebooks. Several of its revisions reuse their wording, and they are credited in Section 6. The NEXUS Runtime Specification and its Reality Mapping companion were drafted by Perplexity at the author's instruction and are cited as specifications. A review of the draft in a separate Claude session raised the monotone-systems objection to the delayed-ring cycle and proposed the horizon test reported in Section 10.1.

Appendix A. The resolver source

The loop-closure root as executed in REG cell 1 [I]. `resolver.py` and the copy in `run_checks.py` carry the same function bodies; a direct comparison found lines 9–39 of `resolver.py` identical to lines 26–56 of `run_checks.py`, and REG cell 1 differs from them only in whitespace and comments.

```
import numpy as np
from scipy.optimize import brentq

# ======================================================================
# ROOT 1 -- LOOP CLOSURE.
# One condition:  sum_k arg H_k(i w) + w*tau = Phi   and   G * prod_k |H_k(i w)| = 1
# A "class descriptor" is a list of crossings, a delay, and a phase branch.
# Crossing kinds:
#   ('lag',  gamma)     first-order     1/(1+s/gamma)
#   ('res',  (wn, zeta))  second-order    wn^2/(s^2 + 2 zeta wn s + wn^2)
#   ('diff', (l, gamma))  diffusive       exp(-l sqrt(s+gamma))
#   ('tele', (L, c, a))   telegrapher     exp(-L sqrt(s(s+2a))/c)
# ======================================================================
def phase_atten(crossings, w):
    """Total phase lag and total log-attenuation of the cascade at frequency w."""
    ph = 0.0; at = 0.0
    for kind, p in crossings:
        if kind == 'lag':
            gam = p
            ph += np.arctan2(w / gam, 1.0)
            at += 0.5 * np.log1p((w / gam) ** 2)
        elif kind == 'res':
```

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```
        wn, z = p; r = w / wn
        ph += np.arctan2(2 * z * r, 1.0 - r * r) % (2 * np.pi)
        at += 0.5 * np.log((1 - r * r) ** 2 + (2 * z * r) ** 2)
    elif kind == 'diff':
        l, gam = p; rr = np.hypot(gam, w)
        ph += l * np.sqrt((rr - gam) / 2)       # analytic branch: monotone, unbounded
        at += l * np.sqrt((rr + gam) / 2)
    elif kind == 'tele':
        L, c, a = p; z = -w * w + 2j * a * w
        k = np.sqrt(np.abs(z)) / c; th = np.angle(z)
        ph += L * k * np.sin(th / 2)
        at += L * k * np.cos(th / 2)
  return ph, at

def resolve(crossings, tau=0.0, Phi=np.pi):
  """THE function. Returns (G_crit, w*), or (None, None) if the phase bill is unpayable."""
  if Phi == 0.0:                    # zero-frequency branch: magnitude alone
    return float(np.exp(phase_atten(crossings, 0.0)[1])), 0.0
  F = lambda w: phase_atten(crossings, w)[0] + w * tau - Phi
  if F(1e12) < 0:
    return None, None
  hi = 1.0
  while F(hi) < 0:
    hi *= 2
  w = brentq(F, 1e-13, hi, xtol=1e-15, rtol=1e-15)
  return float(np.exp(phase_atten(crossings, w)[1])), w
```

The Gram-positivity root as executed in REG cell 5 **[I]**:

```
def gram_gap(x, p):
  """Smallest eigenvalue of X1 - p X2 for two states of overlap x. >= 0 means admissible."""
  X1 = np.array([[1.0, x], [x, 1.0]])
  X2 = np.array([[1.0, x*x], [x*x, 1.0]])
  return np.linalg.eigvalsh(X1 - p * X2).min()
```

Three properties of the code bear on the claims in the text. The $\Phi = 0$ branch returns the magnitude at $\omega = 0$ without root-finding, which is why row 1 of Table 3 holds by construction. The refusal `return None, None` fires when the total phase at $\omega = 10^{12}$ still falls short of Φ, which is how row 9 refuses. The diffusive branch computes phase on the analytic branch, the fix for error E12. The telegrapher branch uses `np.angle` on $z = -\omega^2 + 2i a\omega$, which is safe because z stays in the closed upper half-plane for $\omega > 0$ and $a \geq 0$.

Appendix B. Transcript of the executed checks

Environment and summary of `run_checks.py` (598 lines; output `run_checks.out`, 352 lines; total runtime 7.8 s). Blocks are verbatim; "…" marks lines omitted here, and the full output is in the file.

```
python 3.11.15  numpy 2.4.4  scipy 1.17.1  sympy 1.14.0
SUMMARY: 1:PASS  2:PASS/PASS/FAIL  3:PASS  4:PASS  5:PASS  6:PASS  7:PASS  8:PASS  9:PASS
```

Check 1: the resolver re-run.

```
# class descriptor        crossings kind(params)  tau Phi       w*     G_crit raw  reported literature  rel.err |G H(iw*)-e^-iPhi| match
1 2 lags, Phi=0          2 x lag(1)          0  0        0   1.0000000000  1.000000  1.000000 0.0e+00        0.0e+00 True
2 Hill toggle n=2, Phi=0   (closed form only)      -  0        -          -  2.000000  2.000000 0.0e+00           - True
3 3 lags, Phi=pi         3 x lag(1)          0  pi   1.7320508   8.0000000000  2.000000  2.000000 0.0e+00        2.3e-16 True
4 5 lags, Phi=pi         5 x lag(1)          0  pi   0.72654253   2.8854382000  1.236068  1.236068 2.2e-16        2.5e-16 True
```

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```
5 1 lag + resonator z=0.01  lag(1) + res(1,0.01)     0  pi    1.0099505    0.0404000000  0.040400  0.040400 4.4e-16
5.7e-16 True
6 pure diffusive crossing   diff(1,0)            0  pi    19.739209    23.1406926328 23.140693 23.140693 4.4e-16        4.4e-
16 True
7 telegrapher, a=0       tele(1,1,0)          0  pi    3.1415927    1.0000000000  1.000000  1.000000 2.2e-16        2.2e-16
True
8 telegrapher, a=1e6     tele(1,1,1e+06)        0  pi 9.8696044e-06   23.1406926325 23.140693 23.140693 1.3e-11
2.4e-12 True
9 2 lags, Phi=pi         2 x lag(1)           0  pi       none         none     none     none     -           - True
  9 of 9 rows match their literature value (relative tolerance 1e-6, as in the notebook).
  Row 2 does not call resolve(): the notebook evaluates n(n-1)^-((n+1)/n) inline and compares it with 2.
  Routed through resolve(): the Phi=0 branch gives G_crit = 1.0; solving loop gain [n u^n/(1+u^n)]^2 = G_crit
  on the toggle's symmetric steady state u(1+u^n) = a (n=2) gives a = 2.000000000000.
  Row 9 cross-check: roots of (1+s)^2 + G for G in [1e-3, 1e8]: max real part = -1.000000 (never reaches 0).
  Same 2-lag class plus delay tau=0.1: resolve() returns G_crit = 20.671069 at w* = 4.435208 (bound is class-relative).
>>> CHECK 1: PASS -- 9/9 rows reproduce; row 2 is a closed form, reproduced via resolve() only with the toggle map
```

Check 2: closed forms, the exchange rate, and the toggle threshold.

```
sec(pi/3)                    2.000000000000        2  4.4e-16 True
sec(pi/5)                    1.2360679775oo    1.236068  2.3e-08 True
sec(pi/5) - (sqrt5 - 1)          -0.000000000000         0  2.2e-16 True
sec^3(pi/3)  [Griffith n > 8]       8.000000000000         8  5.3e-15 True
e^pi                        23.140692632779  23.1406926  3.3e-08 True
4 zeta(1+zeta), zeta=0.01          0.040400000000     0.0404  0.0e+00 True
acos(1/G)/sqrt(G^2-1), G=2          0.6045997880780      0.6046  2.1e-07 True
 same minus pi/(3 sqrt3)          0.000000000000         0  0.0e+00 True
 N       tau    G per stage   N acos(1/G) + tau sqrt(G^2-1)
 3  0.000000000  2.0000000000            3.141592653590
 2  0.604599788  2.0000000000            3.141592653590
 1  1.209199576  2.0000000000            3.141592653590
  a_c(2) = 2.0000000000
  a_c(3) = 1.1905507890
  a_c(4) = 1.0131142475
2  1.98000  0.990 |  1     1 -0.0050                 |      1 (0.9950,0.9950)                  monostable
2  2.02000  1.010 |  3     2 -0.0099 +0.0050 -0.0099       |      2 (0.8682,1.1518) (1.1518,0.8682)          bistable
2  3.00000  1.500 |  3     2 -0.3333 +0.1911 -0.3333       |      2 (0.3820,2.6180) (2.6180,0.3820)          bistable
...
  Requested statement 'a_c(2) = 4': already bistable (two stable states, both methods) at a = [2.02, 2.1, 3.0, 3.9], all < 4.
  Formula, fixed-point stability and simulation agree on a_c(2) = 2.  (4 = n^2 is only the a -> infinity
  supremum of the n=2 loop gain [n u^n/(1+u^n)]^2; it is not a bifurcation value of a.)
>>> CHECK 2 (closed forms + exchange rate): PASS
>>> CHECK 2 (a_c(n) formula, n=2,3,4, incl. a_c(2)=2, by fixed points + simulation): PASS
>>> CHECK 2 (requested value a_c(2)=4): FAIL -- simulation shows bistability for 2 < a < 4
```

Check 3: Griffith's 8 in the three-stage Goodwin loop.

```
   n  a_H (eigenvalues)  a_H closed form      K at a_H  eigenvalues at a_H
 8.5     23.5564637627   23.5564637627  8.0000000000  -3.00e+00+0.0000000i  -1.67e-16-1.7320508i  -1.67e-
16+1.7320508i
  10      5.7434917750    5.7434917750  8.0000000000  -3.00e+00+0.0000000i  -2.22e-16-1.7320508i  -2.22e-16+1.7320508i
  16      2.0000000000    2.0000000000  8.0000000000  -3.00e+00+0.0000000i  +2.22e-16-1.7320508i  +2.22e-
16+1.7320508i
...
  n = 8   sup K = 7.999999999828   max Re(eig) = -7.181e-12   (formula -1 + K^(1/3)/2 = -7.181e-12)
     a       n_min(a)   n_min - 8  max Re eig at 0.999 n_min  at 1.001 n_min
     2  16.000000000000  8.000e+00           -3.334e-04     3.332e-04
    10   9.131268655356  1.131e+00           -3.424e-04     3.422e-04
```

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```
  100   8.134427021280  1.344e-01          -3.359e-04     3.357e-04
 1e+12   8.000000000172  1.724e-10          -3.334e-04     3.332e-04
...
10   6.03067  1.05  4000     3.525e-01    3.525e-01  3.6327  OSCILLATES (oscillate)
  onset period 2 pi/sqrt3 = 3.6276
>>> CHECK 3: PASS
```

Check 4: nepers equal radians; the diffusive floor and its decay form.

```
4001 log-spaced frequencies w in [1e-8, 10^5.5]; phase lag reaches 397.64 rad
  max |atten - lag|/sqrt(w/2) = 2.97e-12;  max |atten - sqrt(w/2)|/sqrt(w/2) = 2.97e-12;  max |lag - sqrt(w/2)|/sqrt(w/2) =
8.01e-16
   19.7392   3.141592654   3.141592654 1.000000000   -3.141593
      32   4.000000000   4.000000000 1.000000000    2.283185
  phase = pi at w* = 2 pi^2 = 19.7392088022; there |H| = e^-pi so G_crit = 1/|H| = 23.1406926328; resolve(): 23.1406926328
at w* = 19.7392088022; e^pi = 23.1406926328
  H(s) = exp(-l sqrt(s + gam)),  l = L/sqrt(D),  Da = gam L^2/(D pi^2) = gam l^2/pi^2
 1.0   1.0   9.86960   27.91545680   27.91545680     85.019695223     85.019695223     85.019695223  0.0e+00
 1.0  10.0   98.69604   65.46754926   65.46754926   33506.143065592   33506.143065592   33506.143065592  0.0e+00
  G_crit depends on Da only (l drops out at fixed Da); Da = 0 is the floor e^pi.
>>> CHECK 4: PASS
```

Check 5: Robertson and Robertson–Schrödinger as Gram positivity.

```
10000 trials; d uniform in 2..8; A, B random Hermitian; psi random pure state; u = (A-<A>)psi, v = (B-<B>)psi
  violations at relative tolerance 1e-12:  det < 0: 0   Gram eigenvalue < 0: 0   Robertson: 0   Robertson-Schroedinger: 0
state                              Dx       Dp       Dx*Dp  RS bound sqrt(1/4+cov^2)       Im<xp>    exact
Gaussian, sigma=1.3                1.3000000  0.3846154  0.5000000000          0.5000000000  0.50000000000  0.500000
Gaussian, displaced x0=2, boosted p0=3     1.3000000  0.3846154  0.5000000000          0.5000000000  0.50000000000
0.500000
chirped Gaussian exp(-x^2/4 + 0.3i x^2)    1.0000000  0.7810250  0.7810249676          0.7810249676  0.50000000000
0.781025
two-Gaussian superposition, centres +-3    3.1466041  0.4746360  1.4934917497          0.5000000000  0.50000000000
-
smoothed square pulse |x|<4, edge 0.5      2.1972344  0.4216370  0.9264353650          0.5000000000  0.50000000000      -
x * Gaussian (sigma=1)             1.7320508  0.8660254  1.5000000000          0.5000000000  0.50000000000  1.500000
>>> CHECK 5: PASS
```

Check 6: Lawvere's diagonal on finite sets.

```
case                  |A|  |Y^A|  #e enumerated  #point-surjective e  #e with d in image(e)
Y={0,1},  sigma = negation    3    8        512             0             0
Y={0,1,2}, sigma = y+1 mod 3    3    27       19683             0             0
...
case                    1   2   3   4   5   6   7   8   9  10  11  12
Y={0,1},  sigma = negation      0   0   0   0   0   0   0   0   0   0   0   0
Y={0,1,2}, sigma = y+1 mod 3     0   0   0   0   0   0   0   0   0   0   0   0
control: Y={0,1}, sigma = id   5000 3748 2876 2061 1359  890  527  341  164   88   58   26
(C) Y = {0}: Y^A has a single element, so every e is point-surjective; the only sigma (identity) has a fixed point.
>>> CHECK 6: PASS
```

Check 7: the Gisin–Massar surface, Laplace's rule, and the Duan–Guo boundary.

```
  F(1,2) = 5/6  (== 5/6: True);   F(N,N) == 1 for N = 1..200: True
  sympy: lim M->oo F = (N + 1)/(N + 2);   F - (N+1)/(N+2) = N/(M*(N + 2));   F(N,N) = 1
  26 (N,M) pairs, N = 1..4, M = N..8:  max |numeric - formula| = 6.7e-16
  N   Int p (N+1)p^N dp   Beta(N+1,1).mean  (N+1) Int t^(N+1) dt   (N+1)/(N+2)  max|diff|
  1  0.666666666666667 0.666666666666667    0.666666666666667 0.6666666667   1.1e-16
 200  0.995049504950494 0.995049504950495    0.995049504950497 0.9950495050   1.9e-15
```

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```
  Haar check, 200000 random qubits: t mean 0.50004 (1/2), var 0.08321 (1/12 = 0.08333), KS p-value vs U(0,1) = 0.939
  1000 x in [0, 0.999]: max |p_max - 1/(1+x)| = 6.55e-15;  feasible at p_max-1e-9 and infeasible at p_max+1e-9: True
>>> CHECK 7: PASS
```

Check 8: Gauss flux and $1/r^{d-1}$.

```
d k  unit area (quad)  2pi^(d/2)/G(d/2)     flux r=1    flux r=10  max/min-1 over r  max rel err  r-indep.  k=d-1
2 1   6.283185307180   6.283185307180  6.283185307 6.283185e+00      2.220e-16    2.2e-16 True    True
3 2  12.566370614359  12.566370614359 12.566370614 1.256637e+01      4.441e-16    4.6e-15 True    True
4 3  19.739208802179  19.739208802179 19.739208802 1.973921e+01      2.220e-16    8.4e-15 True    True
d k   max |div F|   max |div F - (d-1-k) r^-(k+1)|  div-free
3 2    3.025e-10                3.025e-10  True
3 3    7.076e-01                2.512e-10  False
>>> CHECK 8: PASS
```

Check 9: the no-cloning corner.

```
X1 - 1*X2 = [[0, -x**2 + x], [-x**2 + x, 0]];  eigenvalues = [-x*(x - 1), x*(x - 1)]
lambda_min = -|x - x^2| >= 0  <=>  x - x^2 = 0  <=>  x in {0, 1}  (a two-point set)
numeric: gram_gap(x, 1) on 100001 points of [0,1]: feasible x = [0.0, 1.0];  max |gram_gap(x,1) + |x - x^2|| = 0.0e+00
complex overlaps x = r e^(i theta) on a 201 x 360 polar grid of the closed unit disk: feasible = [0j, (1+0j)]
  so p_max(1) = 1, while 1/(1+x) -> 1/2 as x -> 1-: the Duan-Guo rate 1/(1+x) holds on [0,1) and jumps at x = 1.
>>> CHECK 9: PASS
```

Appendix C. Notebook and source index

The programme's executed notebooks are in **/home/claude/out/**, with run logs **run.txt** to **run11.txt** beside them. Upload files are in **/root/.claude/uploads/8b6f1298-b896-5271-b2b0-ed60247ac4b1/**.

Table C.1. The eight executed notebooks of the loop/channel programme.

Abbrev.	File	Executed	Question	Headline result
VRT	verb_requirement_test.ipynb	2026-09-25 09:21	what must a reaction network contain to latch and to count?	$a_c(n)$ 9/9; sec(π/N) 5/5 from the one parity miss
OB	one_bill_four_currencies.ipynb	2026-09-25 11:32	break the verb test's symmetric assumptions	3000/3000; master law; preemption; M_c
LOAN	a_loan_not_a_currency.ipynb	2026-09-25 13:29	is noise a currency or a loan?	CV slope −0.497; Γ pinned at Re λ; blind threshold
PRICE	the_price_of_a_crossing.ipynb	2026-09-25 14:34	stage count or dynamical order; does crossing kind matter?	$4\zeta(1+\zeta)$ 10/10; order bound
CEIL	a_price_or_a_ceiling.ipynb	2026-09-25 19:31	are there requirements no resource moves?	e^{π} floor; telegrapher sweep; 5/6 optimum
SIG	the_ceiling_is_in_the_signature.ipynb	2026-09-26 09:53	is the ceiling in the verb or its specification?	ceiling = conjunction; $1/(1+x)$ 7/7
ARROW	the_arrow_and_the_coordinate.ipynb	2026-09-26 11:28	resource vs class extension, measured	frontier shift; Choi rank; TP ⊊ TNI
REG	the_registry.ipynb	2026-09-26 12:10	are named results packages resolved by one interface?	9 rows; 18 packages, 4 roots; empty slot

Figures 1 to 8 are, in order, **fig1_address**, **fig3_phase_closure**, **fig2_transport_axis**, **fig4_copy_frontier**, **fig8_uncertainty**, **fig5_registry**, **fig7_local_global** and **fig6_chain** in **/home/claude/out/paper_figures/**; they are numbered here by first appearance. Scripts and data: **resolver.py** and **root2.py** (the registry's precursors);

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setcomp.py (the six-point set-completion test); **global1.py** (the 8-history global run); **paper_checks/horizon_test.py** with **horizon_test.out** and **horizon_test.json** (the horizon test of Section 10.1); **paper_checks/run_checks.py** and **run_checks.out** (Checks 1–9); **paper_figures/fig1_address** through **fig8_uncertainty** (**.py** and **.png**), with **fig7_rerun.py**, **fig7_rerun_data.json** and **fig7_global1_rerun.txt** for the 96-history re-run.

The corpus is **64956014-MergedNotebook.md** (MN, 109,603 lines). Its sections cited here are:

- arena §8, the ε closed form and two kills (L322–1094);
- the ε sweep (L1095–1486);
- §9, the reader bridge (L1487–1894 live);
- §10, substrate change (L1–321);
- §11, subdivision (L2154–2444);
- §12, contraction and exclusion (L108555–109604);
- §13, the capacity defect (L108201–108554);
- "Transformation without loss" (L78446–79299);
- the invariant harmonic sector (L12795–14073);
- "Two Quotients of One Machine" v1.1 (L80505–81894);
- "Where the Carry Bit Comes From" v5 (L104826–106849);
- "What a Cut Must Do to Make a Thing" (L99074–100653);
- "Carry Capacity and the Conditional Cut Law" (L7015–8334);
- "The NEXUS Constraint Chain — Executable Record" (L19772–24472);
- the constraint cascade (L84567–85323).

Specifications and reviews:

- the runtime specification: **9fa4a2c6-attachment.txt**, NEXUS Runtime Specification v0.1;
- the reality mapping: **345c851b-attachment.txt**;
- the author's documentation prompts: **1d679eda-attachment.txt**;
- the constraint-chain review: **cbe80e2a**;
- the Co Audit Trail review: **d6c66c0c**;
- the loop-programme reviews: 328aa206, 113d86ac, 7d85f9b2, 996e9c2a, c15e8cb5, 7087f74c, ab9ffd4a, 29456a8b, bc4a302a, c7f1ed92, 1b2d726a, 6c4b99f7, 379599a2, c5b54906, 718a93ab, 2957aa2e and ea6c71ae.

Chat exports:

- NTS, a Perplexity thread, 16–26 September 2026;
- MHC, "Much has changed" (Claude), 21–25 September 2026;
- VAS, "Verbs across substrates" (Claude), 25–26 September 2026.

Earlier documents, not promoted: **bb0f9983** ("Gödel Curvature and Harmonic Recursion", December 2025) and **68b10c6c** ("Quantum Cut-Density as the Origin of Spacetime Curvature", April 2026).

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Appendix D. Glossary of frozen terms

Table D.1. Glossary of frozen terms.

Term	Frozen meaning in this paper
address	the tuple (root ρ, class or signature Σ, coordinate θ, reader R) at which a named result is a boundary value
root	a test of admissibility written as one condition: loop closure, Gram positivity, dilation, counting measure; the diagonal is a candidate
class / signature Σ	the family of admissible transformations: a crossing list, or a task with its flags
coordinate θ	a parameter locating a point inside a class (γ, ζ, ℓ, τ, Φ, N, x, p, (N, M), d, Ω)
reader R	the map from a solution to the published number: G_{crit}, per-stage gain, admissibility, a marginal, a flux, a representation map
resolver	the map from an address to its boundary value; implemented by **resolve()** and **gram_gap()**
named law	a coordinate bound inside a named class; its name records who first compiled the class
package, registry	a named law filed under its root; the table of all packages with their roots
empty slot	local → global: the registry column with no executed package in either corpus family; for sign-structured loops, order structure is the first candidate to pass a test
coordinate move (price)	a change of θ inside a fixed class; the boundary moves; $T_{E(\Sigma)} = T_{\Sigma}$
class extension (arrow)	a change of Σ with $T_{E(\Sigma)} \supsetneq T_{\Sigma}$; opens coordinates
reader change	identification or hiding; changes what is visible without changing T_{Σ}
floor	a coordinate exhausted inside a class that a class extension can escape (e^{π})
ceiling	a coordinate exhausted inside a specification whose conjuncts are each a class extension away (5/6)
gate; bill; currency	retired vocabulary from PRICE and OB; do not use
loan	a coordinate that borrows amplitude at small size and repays at the damping rate (noise)
branch Φ	the total phase a loop must close: 0 latch, π clock, 2π preempted
preemption	a real (latch) crossing opens before any complex crossing on a positive loop
order bound	no finite Hopf threshold at total dynamical order 2 in delay-free, non-self-activating loops
exchange rate	$d\tau/dN = -\arccos(1/G)/\sqrt{G^2-1}$, a map between two classes that reach the same phase
transport axis	crossing kinds ordered by magnitude paid per radian: delay (none), lag, diffusion (one neper), resonance (returns gain)
set completion	the smallest environment dimension at which the attainable set of channels stops growing
Choi rank	rank of $J(\Phi)$: minimal number of Kraus operators, and minimal Stinespring environment
TP, TNI	trace-preserving; trace-non-increasing (heralded) operations
finite-horizon occupancy Occ(τ, T)	the share of sampled histories still oscillating at run length T; not a basin fraction unless it is stable as T grows
attractor atlas	the specified but unbuilt map of attractors, their regions and basins
EXTEND, HIDE, REFACTOR, IDENTIFY, EXCLUDE	the runtime's meta-operations; only EXCLUDE removes admissible continuation

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Term	Frozen meaning in this paper
C_0, C_1, C_{1a}, C_2, C_3	C_0: distinction exists; C_1: all things must change (continuation is required); C_{1a}: independent continuation; C_2: history preserved through transitions (the C-chain used in the Keyes mapping) or infinite degrees of freedom for the potential to change (the constraint chain); C_3: all change is equal
Law 2	no matter may hold a location without the potential to be moved from that location
Law 3	there is no end (cascade); not to be merged with the arena or recurrence candidates
hold tax	$\min\langle p^2\rangle/2 = 1/(8\sigma_x^2)$, the diverging price of holding a location
FOLD, FIX, Ω_3	an involution that returns; a behavioural fixed point $P \simeq T(P)$; the open question of whether FIX is irreducible
audit the reader	before revising a model, ask which quotient, branch, marginal, measure or concentrated region the measurement discarded

Appendix E. References

Each entry states how far it was verified in this research. **[V]** means the content was read. **[V-c]** means only the bibliographic record was confirmed. DOIs marked "unverified" come from bibliographic knowledge and must be checked before print.

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The NEXUS corpus (CC BY-NC 4.0; repository github.com/QuHarmonics). The eight programme notebooks, the merged corpus notebook, the runtime specification and the placement checks are indexed in Appendix C.

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